Double Spacetime Theory: Gravity as Torsion between Usual and Dual Spacetime

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Abstract

General Relativity has taught us to see gravity as the curvature of a single spacetime continuum. That teaching accounts for the motion of planets, the delay of light, and the ringing of black holes. It does not say why a continuum should curve, nor why the same phenomenon should appear as inertia under acceleration. The present work answers those questions by refusing the continuum as a primitive. Spacetime is not an autonomous stage. Each massive particle carries a pair of Minkowski frames—the usual, in which we live and measure, and a dual, time-reversed relative to the first, with the same interval. Gravity is the torsion of that pair: a mismatch that belongs to the particle itself.

From this picture two things follow at once. The geometry outside a spherical mass is recovered as a mismatch of the two frames, without taking curvature as a primitive. Gravity and inertia become the same phenomenon: free fall is the state in which the two frames stay aligned, and acceleration is the same misalignment when it is sourced by surrounding mass. If the ontology is sound, then the geometry of the world is not prior to matter. It is an attribute of particles, as spin and charge are attributes, and the apparent curvature of the heavens is the torsion of each particle's own dual frames.

1 Introduction

Since 1915, general relativity has been the most successful description of gravitation we possess. The perihelion of Mercury, the delay of light, gravitational waves, and the imaging of black holes all speak for it. Yet a philosophical unease remains: why does matter curve spacetime, and why does curved spacetime dictate the motion of matter? The Einstein equations tell us how gravity works with exquisite precision. They do not say why a continuum should curve at all. The unease sharpens against Mach's principle. The inertia of a body ought to be determined by the totality of matter in the universe, yet general relativity treats spacetime as an independent continuum that can exist even in complete emptiness.

A century of attempts to close this gap has left the continuum itself untouched. Spacetime is taken to be a smooth manifold a priori, and gravity is the metric's deviation from flatness.

The present work refuses that primitive. There is no single shared spacetime. Each massive particle carries its own pair of Minkowski frames. One is the usual spacetime, in which we live and measure. The other is a dual spacetime: time-reversed relative to the first, with the same interval. Gravity and inertia are not curvature of a background. They are the mismatch of these two frames when one particle is asked to stay parallel to another.

The shift is ontological. In the standard picture, spacetime tells matter how to move and matter tells spacetime how to curve. Here there is no autonomous spacetime at all. The degrees of freedom we call spacetime are internal attributes of particles, as spin and charge are attributes. The apparent curvature of the macroscopic world is that same mismatch, belonging to each particle.

Two consequences follow at once, and they can be stated before any formula is written. First, the geometry outside a spherical mass is recovered as a mismatch of the two frames, without taking curvature as a primitive. Second, gravity and inertia are the same phenomenon. Free fall is the state in which the two frames stay aligned. Acceleration breaks that alignment, and the resulting mismatch is the inertial force—the same torsion that we call gravity when it is sourced by surrounding mass.

This is not a modification of general relativity. It is a proposed completion: the continuum remains a useful macroscopic fiction, while spacetime degrees of freedom are treated as paired and particle-local.

The paper proceeds in that order. Sections 2 and 3 introduce the two sectors and the transformations that move them. Section 4 defines gravity as their mismatch and recovers the exterior of a spherical mass. Section 5 reads inertia from the same mismatch, so that the equivalence principle is no longer a

postulate. With that picture in hand, later sections take the geometry to strong fields, to nuclear and atomic scales, to electromagnetism, and to galaxies. Those readings are built on the picture; they are not its foundation. Section 18 returns to the ontology and to the observations that would test it.

2 Mathematical Foundation: The 16-Dimensional Biquaternion Algebra

The entire double spacetime theory is constructed within the 16-real-dimensional algebra of biquaternions (also known as double quaternions), which is canonically isomorphic to the Clifford geometric algebra $\mathrm{Cl}(3,1)$ over Minkowski spacetime with signature $(-,+,+,+)$. This algebra naturally accommodates two complete, mutually commuting copies of Minkowski spacetime — the usual and the dual — within a single algebraic structure intrinsically attached to every massive particle.

2.1 The Biquaternion Algebra

We begin with two independent copies of Hamilton's quaternion algebra:

- Primary quaternions generated by i, j, k obeying $i^2 = j^2 = k^2 = -1$, $ij = k$, $ji = -k$ (cyclic),
- Secondary quaternions generated by I, J, K obeying identical rules $I^2 = J^2 = K^2 = -1$, $IJ = K$, $JI = -K$ (cyclic).

The full biquaternion algebra is their tensor product $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H}$, in which the two sets strictly commute:

$$iI = Ii,\ iJ = Ji,\ iK = Ki,\ jI = Ij,\ \ldots$$

The resulting 16 linearly independent real basis elements are

$$1,\ i,\ j,\ k,\ I,\ J,\ K,\ iI,\ iJ,\ iK,\ jI,\ jJ,\ jK,\ kI,\ kJ,\ kK.$$

2.2 Explicit Isomorphism with Cl(3,1)

The algebra is isomorphic to $\mathrm{Cl}(3,1)$ via the following identification of grade-1 (vector) generators:

$$e_0 = j, \qquad \text{(time-like basis vector)}$$

$$e_1 = kI, \quad e_2 = kJ, \quad e_3 = kK \qquad \text{(space-like basis vectors)},$$

which immediately satisfy the defining relations

$$e_0^2 = -1, \quad e_1^2 = e_2^2 = e_3^2 = +1, \quad \{e_\mu, e_\nu\} = 0 \ \ (\mu \neq \nu).$$

The higher-grade elements then follow standard Clifford multiplication. In particular:

$$\begin{aligned}
&\text{grade-2 (bivectors):} && e_0e_1 = iI,\ \ e_0e_2 = iJ,\ \ e_0e_3 = iK,\\
&&& e_3e_2 = I,\ \ e_1e_3 = J,\ \ e_2e_1 = K,\\
&\text{grade-3 (trivectors):} && e_1e_2e_3 = k,\ \ e_0e_3e_2 = jI,\ \ e_0e_1e_3 = jJ,\ \ e_0e_2e_1 = jK,\\
&\text{pseudoscalar:} && e_0e_1e_2e_3 = i.
\end{aligned}$$

The complementary "dual" basis (time-reversed under right-multiplication by i) is generated by

$$\tilde{e}_0 = k = e_1e_2e_3, \quad \tilde{e}_1 = jI = e_0e_3e_2, \quad \tilde{e}_2 = jJ = e_0e_1e_3, \quad \tilde{e}_3 = jK = e_0e_2e_1,$$

yielding the dual spacetime representation introduced below.

This explicit isomorphism makes manifest that the 16-dimensional biquaternion algebra is not an ad-hoc construction but the geometrically natural Clifford algebra of 4-dimensional Minkowski spacetime, now enriched with an intrinsic duality that distinguishes usual and dual sectors while preserving the full Lorentzian structure.

The apparent irregularity in the Clifford algebraic forms — such as the asymmetric placement of j as the sole time-like vector generator and the cross-term-dominated spatial basis kI, kJ, kK — is not accidental: it exists precisely to preserve the full rotational symmetry of three-dimensional physical space within a structure that simultaneously encodes two time-reversed copies of Minkowski spacetime. This subtle asymmetry is what allows the dual map $X \mapsto Xi$ to reverse the arrow of time without breaking spatial isotropy, thereby providing the algebraic foundation for both the equivalence principle and the emergence of gravitational torsion as a relative misalignment between particle-intrinsic dual frames.

2.3 Dual Spacetime Structure Intrinsic to Particles

Each particle is postulated to carry an intrinsic pair of compactified Minkowski spacetimes encoded within the biquaternion algebra:

- **Usual Spacetime:** Represented by vectors of the form

$$X = ct\, j + x\, kI + y\, kJ + z\, kK,$$

  with Minkowski norm

$$X^2 = -(ct)^2 + x^2 + y^2 + z^2.$$

- **Dual Spacetime:** Represented by vectors of the form

$$X' = ct'\, k + x'\, jI + y'\, jJ + z'\, jK,$$

  with identical Minkowski norm

$$X'^2 = -(ct')^2 + x'^2 + y'^2 + z'^2.$$

The two spacetimes are related by the dual map

$$X' = Xi,$$

which reverses the arrow of time:

$$(ct\, j)i = ct\,(ji) = -ct\, k,$$

while preserving the Minkowski norm. Since the pseudoscalar $i = e_0e_1e_2e_3$ of $\mathrm{Cl}(3,1)$ anticommutes with every grade-1 vector ($iX = -Xi$ for $X \in \mathrm{Cl}(3,1)$) and satisfies $i^2 = -1$, the norm transforms as

$$X'^2 = (Xi)(Xi) = X(iX)i = -X(Xi)i = -X^2\, i^2 = -X^2 \cdot (-1) = X^2.$$

This dual structure is intrinsic to every particle, with the usual spacetime governing its standard kinematics and the dual spacetime encoding complementary degrees of freedom that become dynamically relevant in the presence of gravity and inertia. The existence of these two spacetimes allows for a richer geometric interpretation of Lorentz transformations, parallel transport, and ultimately the nature of gravitation itself.

3 Kinematic Structure: Rotors, Boosts, and Parallel Transport

In this section we develop the kinematic framework of the theory. Lorentz boosts of the usual sector and elliptic rotations of the dual sector are generated by the six bivectors $\{i\Gamma_a, \Gamma_a\}$. Same-axis pairs commute; off-axis pairs need not. Boosts arise from bivectors that square to $+1$, yielding the cosh/sinh expansion of the rotor.

3.1 Rotors in the Usual Spacetime

To illustrate the kinematic structure, consider a Lorentz boost along the x-direction with rapidity ϕ. The generating bivector is iI, which satisfies $(iI)^2 = i^2I^2 = (-1)(-1) = +1$. In general, the usual-sector boost generators iI, iJ, iK square to $+1$, while the dual-sector rotation generators I, J, K square to -1. The same I, J, K are the spatial bivectors of the usual Minkowski frame ($e_3e_2 = I$, $e_1e_3 = J$, $e_2e_1 = K$); they do not furnish a second, independent copy of $\mathfrak{so}(3)$.

The corresponding boost rotor is thus

$$R_{\text{usual}} = \exp\left(\frac{\phi}{2} iI\right).$$

Given the positive square of the generator, the exponential expands naturally into hyperbolic functions:

$$R_{\text{usual}} = \cosh\frac{\phi}{2} + iI \sinh\frac{\phi}{2}.$$

Applying this rotor via the sandwich product to a spacetime vector $X = ct\,j + x\,kI + \cdots$ yields the canonical Lorentz transformation:

$$\begin{aligned} ct'\,j + x'\,kI &= R_{\text{usual}}\,X\,R_{\text{usual}}^{\dagger} \\ &= \left[\cosh\frac{\phi}{2} + iI\sinh\frac{\phi}{2}\right](ct\,j + x\,kI)\left[\cosh\frac{\phi}{2} - iI\sinh\frac{\phi}{2}\right] \\ &= (\gamma ct + \gamma vx)\,j + (\gamma x + \gamma vct)\,kI, \end{aligned}$$

where $\gamma = \cosh\phi$ and $v/c = \tanh\phi$. Spatial rotations of the usual Minkowski vectors are generated by the same $\Gamma_a \in \{I, J, K\}$ that generate dual-sector elliptic rotors; they are not an extra Lorentz copy.

3.2 The Complete Rotor

The most general six-parameter exponential on this basis is

$$R_{\text{total}} = \exp\left(\frac{\phi_1}{2}iI + \frac{\phi_2}{2}iJ + \frac{\phi_3}{2}iK + \frac{\theta_1}{2}I + \frac{\theta_2}{2}J + \frac{\theta_3}{2}K\right),$$

where the parameters ϕ_a govern usual-sector boosts and θ_a dual-sector rotations. Write $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$ for the spatial bivectors of Section 2. Factorisation $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$ is not automatic: it requires the two summands to commute. About a single axis they do: $[i\Gamma_a, \Gamma_a] = 0$, so

$$\exp\left(\tfrac{\phi}{2}i\Gamma_a + \tfrac{\theta}{2}\Gamma_a\right) = \exp\left(\tfrac{\phi}{2}i\Gamma_a\right)\,\exp\left(\tfrac{\theta}{2}\Gamma_a\right).$$

About different axes they anticommute. A combination $p\,i\Gamma_a + q\,\Gamma_b$ with $a \neq b$ squares to the scalar $p^2 - q^2$, and the exponential is one rotor, hyperbolic, elliptic, or parabolic according to the sign of that scalar. In particular $[iI, J] \neq 0$, and the six-parameter exponential does not factor into a product of one-axis rotors without that extra assumption. In what follows we work on a single axis whenever we write $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$.

To make the geometric meaning more transparent on a single axis, we now introduce the bivector form of the rotor, which separates magnitudes from directions.

Define the unit boost bivector in the usual spacetime

$$\hat{p} = p_1\,iI + p_2\,iJ + p_3\,iK, \qquad \hat{p}^2 = +1,$$

where $\hat{p}$ is normalized ($|\hat{p}| = 1$) and $\phi = \sqrt{\phi_1^2 + \phi_2^2 + \phi_3^2}$ is the total rapidity (with direction encoded in the components $p_a = \phi_a/\phi$).

Similarly, define the unit rotation bivector in the dual spacetime

$$\hat{q} = q_1\,I + q_2\,J + q_3\,K, \qquad \hat{q}^2 = -1,$$

where $\hat{q}$ is normalized ($|\hat{q}| = 1$) and $\theta = \sqrt{\theta_1^2 + \theta_2^2 + \theta_3^2}$ is the total dual rotation angle (with direction encoded in $q_a = \theta_a/\theta$).

The complete rotor then takes the bivector-exponential form

$$R_{\text{total}} = \exp\left(\frac{\phi}{2}\hat{p} + \frac{\theta}{2}\hat{q}\right).$$

When $\hat{p}$ and $\hat{q}$ are aligned on a single axis the exponential factorises, $R_{\text{total}} = R_{\text{usual}}R_{\text{dual}}$, with

$$R_{\text{usual}} = \exp\left(\frac{\phi}{2}\hat{p}\right) = \cosh\frac{\phi}{2} + \hat{p}\sinh\frac{\phi}{2},$$

$$R_{\text{dual}} = \exp\left(\frac{\theta}{2}\hat{q}\right) = \cos\frac{\theta}{2} + \hat{q}\sin\frac{\theta}{2}.$$

The six real parameters remain the rapidity ϕ and unit boost direction $\hat{p}$ in the usual sector, and the rotation angle θ and unit axis $\hat{q}$ in the dual sector. The factorisation is used below only when the generators commute.

3.3 The Sandwich Product

The versor formalism provides a compact means to apply these transformations: for any biquaternion multivector X representing a spacetime event, the transformed vector is

$$\tilde{X} = R_{\text{total}}\, X\, R_{\text{total}}^{\dagger}.$$

This single operation preserves the Minkowski norm ($X^2 = \tilde{X}^2$) and acts coherently on both usual and dual components, seamlessly integrating boosts and rotations into one algebraic expression. No matrices or coordinate-dependent machinery are required; the geometry unfolds intrinsically within the algebra.

4 Gravity as Torsional Mismatch between Dual Spacetime

In this central section, we establish that gravity is neither spacetime curvature nor an external force, but the torsional mismatch between the usual and dual spacetimes intrinsically carried by every massive particle. Christoffel symbols, Riemann tensors, and a continuous manifold are not primitive objects. Torsion is the relative rotation angle between particle-local dual rotors. The parameter-space scalar J is distinct from the Weitzenböck scalar T.

4.1 Definition of the Torsional Mismatch Rotor

The torsional mismatch rotor is defined as

$$\Omega = R_{\text{usual}}^{\dagger} R_{\text{dual}}.$$

The dagger operation is the standard conjugate-inverse of the rotor in the biquaternion algebra:

$$R_{\text{usual}}^{\dagger} = \exp\left(-\frac{\phi}{2}\hat{p}\right) = \cosh\frac{\phi}{2} - \hat{p}\sinh\frac{\phi}{2}.$$

Although formally a simple conjugation, this dagger carries special significance in the double spacetime framework: the negative sign in front of the sinh term precisely encodes the time-reversal inherent in the dual map. Since the pseudoscalar i is grade-4 and therefore commutes with bivectors, right-multiplication by i acts on the boost generator as

$$\hat{p}\, i = -(p_1 I + p_2 J + p_3 K) \equiv -\hat{q}', \qquad (\hat{p}\, i)^2 = -1,$$

where $\hat{q}' \equiv p_1 I + p_2 J + p_3 K$ is a unit rotation generator in the dual sector. Equivalently, $\hat{p} = \hat{q}'\, i$, and the inverse usual rotor admits the exact rewriting

$$R_{\text{usual}}^{\dagger} = \cosh\frac{\phi}{2} - \hat{q}'\, i\, \sinh\frac{\phi}{2}.$$

Using the standard identities $\cosh(\phi/2) = \cos(i\phi/2)$ and $\sinh(\phi/2) = \sin(i\phi/2)/i$, this admits the analytically continued ($\phi \to i\phi$) trigonometric form

$$R_{\text{usual}}^{\dagger} = \cos\left(\frac{i\phi}{2}\right) - \hat{q}'\, \sin\left(\frac{i\phi}{2}\right),$$

revealing that the dagger operation effectively bridges the hyperbolic geometry of the usual spacetime to the elliptic geometry of the dual spacetime—not as an equality of cosh and cos for real arguments, but as the unique analytic continuation of the boost rotor across the duality $X \mapsto Xi$.

The relative rotor $\Omega = R_{\text{usual}}^{\dagger} R_{\text{dual}}$ realizes this bridge operationally: the dagger temporarily "deconstructs" the hyperbolic boost structure of the usual spacetime (inducing the dual map through its sign reversal), while subsequent multiplication by R_{dual} "reconstructs" the combined object, making the relative misalignment manifest. Through this "deconstruct-then-reconstruct" process, the torsional mismatch between the usual and dual spacetimes is extracted algebraically, and this mismatch constitutes the essence of gravity in the double spacetime framework.

Free fall is the state in which this mismatch vanishes. Unitarity of R_{usual} gives a sharp criterion: $\Omega = 1$ if and only if the two rotors coincide, $R_{\text{dual}} = R_{\text{usual}}$. Anti-synchronization $R_{\text{dual}} = R_{\text{usual}}^{\dagger}$ is a different condition; on a single axis it forces both bivector coefficients to vanish, so a nontrivial pure

boost does not satisfy it. We read mass-energy as the physical cost of deviations from $\Omega = 1$. The rotors themselves are units in the even subalgebra. The logarithm $\Omega_{\text{biv}} = \log \Omega$ takes values in $\mathfrak{so}(3,1)$, and the six generators $\{i\Gamma_a, \Gamma_a\}$ span one Lorentz copy, not the twelve-dimensional sum $\mathfrak{so}(3,1) \oplus \mathfrak{so}(3,1)$.

The associated torsion bivector is

$$\Omega_{\text{biv}} = \log \Omega \in \mathfrak{so}(3,1),$$

the principal branch on a simply connected neighbourhood of the identity. Throughout this paper the regime of small mismatch is $\Omega \approx 1$, where Ω_{biv} is uniquely determined; topologically distinct branches lying outside this neighbourhood are not used in the dynamics presented here.

4.2 Lorentz-Invariant Scalar from Dual Torsion

The Killing form on $\mathfrak{so}(3,1)$ (vector representation) is

$$B(X, Y) = 4 \operatorname{Tr}(XY).$$

Denote the three grade-2 spatial bivectors of §3 collectively by Γ_a ($a = 1, 2, 3$) with $\Gamma_1 = I$, $\Gamma_2 = J$, $\Gamma_3 = K$ (the Clifford identification $e_3e_2 = I$, $e_1e_3 = J$, $e_2e_1 = K$ of §2). The usual-spacetime boost generators are $i\Gamma_a$, while Γ_a generate rotations in the dual sector. Evaluating the Killing form gives $B(i\Gamma_a, i\Gamma_a) = +8$ and $B(\Gamma_a, \Gamma_a) = -8$, reproducing the Lorentzian signature perfectly. Same-axis generators commute, $[i\Gamma_a, \Gamma_a] = 0$; off-axis commutators need not vanish.

Write the half-angle expansion $\Omega_{\text{biv}} = \sum_a \left(\frac{\alpha_a}{2} i\Gamma_a + \frac{\beta_a}{2} \Gamma_a\right)$. The six generators span one copy of $\mathfrak{so}(3,1)$; together with translations the ambient algebra is the Poincaré candidate $\mathfrak{iso}(3,1) \cong \mathfrak{so}(3,1) \ltimes \mathbb{R}^{3,1}$. Then

$$B(\Omega_{\text{biv}}, \Omega_{\text{biv}}) = 2 \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2).$$

The scalar that measures the mismatch throughout this paper is

$$J = \frac{1}{2} \sum_{a=1}^{3} (\alpha_a^2 - \beta_a^2), \tag{1}$$

four times $\frac{1}{16} B(\Omega_{\text{biv}}, \Omega_{\text{biv}})$ under this expansion. We also write $J_{\text{norm}} = \frac{8}{3\pi^2} J$, and $M = \frac{1}{2} \sum_a (\alpha_a^2 + \beta_a^2)$ for the unsigned torsional mass, with the same normalisation $M_{\text{norm}} = \frac{8}{3\pi^2} M$.

The sign of J is the distinction between ordinary attractive gravity and dual-dominated repulsion. When usual-spacetime boosts dominate, $J > 0$ and the force is attractive. When dual rotations dominate, $J < 0$ and the force is repulsive. Swapping the two sectors, $\alpha \leftrightarrow \beta$, sends $J \mapsto -J$ and leaves M invariant.

Three vanishing conditions must not be conflated. If $M = 0$, then $J = 0$: every rapidity vanishes, and this is vacuum. The converse is false. Equal usual and dual rapidities give $J = 0$ with $M > 0$—balanced massive states—and on a single axis those states still have $\Omega \neq 1$. Thus $\Omega = 1$, $J = 0$, and $M = 0$ are three different physical situations. Gravity as free fall is $\Omega = 1$, not the vanishing of J. On one axis, with σ the sum of the rapidities and δ their difference, $J = \frac{1}{2}\delta\sigma$ and $M = \frac{1}{4}(\sigma^2 + \delta^2)$: vanishing δ is the balanced locus, and it is the vacuum only when σ vanishes as well.

On admissible configurations—non-negative rapidities with $\alpha_a + \beta_a \leq \pi/2$ on each axis—one has $|J| \leq \frac{3\pi^2}{8}$ and $|J_{\text{norm}}| \leq 1$, and likewise $0 \leq M_{\text{norm}} \leq 1$. The image of J_{norm} on this cone is the whole interval $[-1, 1]$. Equality $|J_{\text{norm}}| = 1$ holds only for a purely hyperbolic configuration ($\alpha_a = \pi/2$, $\beta_a = 0$ on every axis) or a purely elliptic one ($\alpha_a = 0$, $\beta_a = \pi/2$). A principal-branch condition $|\alpha_a + \beta_a| \leq \pi/2$ without the sign restriction is not enough: the bound fails.

What the bound limits is the mismatch a configuration expresses, not the sum of the bivectors that sources contribute. Those bivectors add in the six-dimensional coefficient space, and the sum does not depend on the order of the sources. The internal sum has no ceiling: the same quadratic J, evaluated on the sum, may be driven past $\frac{3\pi^2}{8}$. The expressed bivector is the admissible configuration closest to that sum, in the squared distance of the six coefficients, taken axis by axis on the triangle $\alpha \geq 0$, $\beta \geq 0$, $\alpha + \beta \leq \pi/2$. Inside the triangle the closest point is the sum itself, so a weak collection of sources expresses the ordinary sum of their bivectors. The same addition is a Faraday increment in this six-space. An increment that leaves the triangle is an internal departure; the mismatch then expressed is the closest admissible point, not the quadratic of the departed coefficients.

On an axis whose coefficients are nonnegative, write σ for their sum and δ for their difference. Once $\sigma \geq \pi/2$ while $|\delta| \leq \pi/2$, the expressed sum is $\pi/2$ and the expressed difference remains δ. Once $\delta \geq \pi/2$

the expressed state locks at the pure boost $(\pi/2, 0)$; once $-\delta \geq \pi/2$ it locks at the pure rotation $(0, \pi/2)$. Further sources of the same kind increase the internal sum and leave the expressed bivector unchanged. Thus $|J|$ of the expressed configuration stays at most $\frac{3\pi^2}{8}$ however large the internal sum becomes.

Aligned sources add rapidities, not values of J. One pure boost of rapidity $\pi/2$ on a single axis, repeated $n \geq 1$ times, has internal rapidity $n\pi/2$ and internal quadratic $n^2\pi^2/8$. For every such n the expressed rapidity is $\pi/2$ and the expressed J on that axis is $\pi^2/8$. For $n \geq 2$ the internal quadratic already exceeds the three-axis ceiling $\frac{3\pi^2}{8}$. Two aligned boosts of rapidity $\pi/4$ remain inside the triangle and express one boost of rapidity $\pi/2$: each source has $J = \pi^2/32$, the expressed sum has $J = \pi^2/8$, and the sum of the two separate values is $\pi^2/16$.

4.3 Schwarzschild Chart and the Distinction among J, J_{field}, and T

On the exterior Schwarzschild domain $r > r_s > 0$ the radial pure-boost rapidity is

$$\varphi(r) = -\tfrac{1}{2}\log\bigl(1 - r_s/r\bigr).$$

The associated sandwich scales reproduce the diagonal tetrad legs:

$$\mathrm{e}^{-\varphi} = \sqrt{A}, \qquad \mathrm{e}^{\varphi} = A^{-1/2}, \qquad A(r) = 1 - r_s/r.$$

The coframe $\bigl(\sqrt{A},\, A^{-1/2},\, r,\, r\sin\theta\bigr)$ induces the Schwarzschild metric $\mathrm{diag}(-A,\, A^{-1},\, r^2,\, r^2\sin^2\theta)$. More generally, a rotor field $R_{\text{total}}(x)$ acts on a fixed reference tetrad $\{\mathring{e}^a\}$ by the sandwich product

$$e^a{}_\mu(x) \;=\; \langle R_{\text{total}}(x)\, \mathring{e}^a\, R^\dagger_{\text{total}}(x)\rangle_{1,\mu},$$

yielding the metric $g_{\mu\nu} = \eta_{ab}\, e^a{}_\mu e^b{}_\nu$. For a coframe $e^a{}_\mu$ the Weitzenböck torsion is $T^a{}_{\mu\nu} = \partial_\mu e^a{}_\nu - \partial_\nu e^a{}_\mu$, and the teleparallel scalar is the quadratic combination

$$T = \tfrac{1}{4}T_{\rho\mu\nu}T^{\rho\mu\nu} + \tfrac{1}{2}T_{\rho\mu\nu}T^{\nu\mu\rho} - T_\rho T^\rho,$$

with indices moved by the spacetime metric. On the diagonal coframe the coordinate torsion consists of the radial derivative of the time leg and the angular anholonomy of the spherical frame. The quadratic evaluates to

$$T = -\frac{2}{r^2}\bigl(A + rA'\bigr). \tag{2}$$

The angular pieces cancel, so T does not depend on θ. The flat spherical frame $A = 1$, $A' = 0$ already has $T = -2/r^2$: flatness of the metric does not set this coframe's teleparallel scalar to zero.

Three quadratic quantities appear on this chart and must not be conflated:

$$J = \tfrac{1}{2}\varphi^2, \qquad J_{\text{field}} = \tfrac{1}{2}(\varphi')^2, \qquad T \text{ as in (2)}.$$

The first is the finite-angle mismatch of the two rotors; the second is a field seed built from $\partial_r\varphi$; the third is the teleparallel Lagrangian density. They are not the same object. A constant nonzero boost of the inertial frame in one Cartesian plane already has $J \neq 0$ with vanishing T, so J cannot be a multiple of T in general.

The separation is not an artefact of freezing the rapidity. Boost the inertial frame in one fixed plane by a rapidity $\varphi(t, x)$ that may vary from event to event. The coframe at each event is that boost, and the torsion sees the profile only through $\partial_t\varphi$ and $\partial_x\varphi$. The induced metric is the Minkowski metric for every profile, and the (t, x) block has determinant one: the coframe is a proper boost of the inertial frame. Einstein's equation is an equation for the metric, so the rapidity drops out of it. The coordinate torsion keeps the derivatives and forgets the rapidity itself,

$$T^t{}_{tx} = \partial_t\varphi, \qquad T^x{}_{xt} = \partial_x\varphi.$$

A varying rapidity is a real torsion. The teleparallel quadratic cancels those derivatives as well, and $T = 0$ at every event, for every profile, while $J = \frac{1}{2}\varphi^2$ stays arbitrary. Torsion may be present, the vacuum equation may hold, and the teleparallel density may vanish, without constraining the mismatch. No pointwise function of J is the teleparallel scalar of a general tetrad.

What follows is the comparison inside one ansatz, not a gauge-invariant law. On the static diagonal coframe the same rapidity rescales the time and radial legs, and the metric changes with it. The curvature calculation for $\mathrm{diag}(-A,\, A^{-1},\, r^2,\, r^2\sin^2\theta)$ gives

$$G^t{}_t = G^r{}_r = \frac{A + rA' - 1}{r^2}, \qquad G^\theta{}_\theta = G^\varphi{}_\varphi = \frac{A''}{2} + \frac{A'}{r}.$$

Comparison with (2) is the identity

$$T + \frac{2}{r^2} = -2\,G^t{}_t. \tag{3}$$

The vacuum equation $G^t{}_t = G^r{}_r = 0$ is the single condition $\frac{d}{dr}(rA) = 1$; the angular components are its radial derivative. On an open ray that condition selects a unique member, $A(r) = 1 - r_s/r$, with $r_s = r(1 - A(r))$. That is the exterior Schwarzschild factor, and on it the Ricci scalar vanishes while

$$T = -\frac{2}{r^2},$$

the same value as the flat spherical frame. The mass drops out of T. It does not drop out of the rapidity. On this chart $\frac{r_s}{2r} \le \varphi(r) \le \frac{r_s}{2rA(r)}$, so $r\varphi \to r_s/2$ at infinity, the finite-angle rapidity tracks the Newtonian potential, and $4r^2J \to r_s^2/2$. There is no corresponding falloff for T, and the two densities do not agree in the weak field: $r^2T = -2$ while $J = \frac{1}{2}\varphi^2 > 0$. As $r \to r_s^+$ one has $A \to 0^+$ and $\varphi \to \infty$, while T remains $-2/r^2$. The classical pathology is the unbounded rapidity, which the admissible cone cuts off in Section 6.2. Repulsive layers are the negative sign of J, in the elliptic sector. They are not a negative value of this T: the attractive vacuum itself has $T < 0$.

4.4 Proposed Action

We propose the action

$$S = \frac{c^4}{16\pi G} \int J\, d^4x$$

as a Lagrangian density in the mismatch scalar J. The integral is algebraic and background-independent. It is not the teleparallel integral of T. For T itself the standard identity

$$\int T\, e\, d^4x = -\int R\,\sqrt{-g}\, d^4x + \int \partial_\mu(\sqrt{-g}\, V^\mu)\, d^4x$$

holds after tetrad variation with vanishing boundary variations, so the Einstein equations belong to T, not to J. Gravity in this theory remains the mismatch angle between particle-intrinsic dual frames. On the vacuum radial chart the tetrad determinant is $e = r^2 \sin\theta$, so $e\,T = -2\sin\theta$, independent of the mass, while J carries the redshift. The two do not agree in the weak field. Off that chart the comparison is empty in a sharper sense: the inertial boost just described has vanishing T for every J, and the spherical frame of Minkowski space has $T = -2/r^2$ with $J = 0$. The Einstein equation constrains the metric. On the diagonal radial ansatz it is $\frac{d}{dr}(rA) = 1$, which freezes T at the spherical-frame value and selects Schwarzschild. It does not select J, and it does not select the rotor of a general tetrad.

4.5 Weak-Field and Newtonian Limit

In the linearized regime,

$$R_{\text{usual}} \approx 1 + \frac{1}{2}\sum \alpha_a\, i\Gamma_a, \quad R_{\text{dual}} \approx 1,$$

yielding $\Omega_{\text{biv}} \approx \sum \alpha_a\, i\Gamma_a$ and $J \approx \frac{1}{2}\sum \alpha_a^2$. If the boost parameters track the Newtonian potential as $\alpha_a \propto \partial_a(\Phi/c^2)$, Poisson's equation

$$\nabla^2\Phi = 4\pi G\rho$$

is recovered with the standard coefficient. This matching identifies G with the linearised chart; it does not derive G from the algebra. On the exterior Schwarzschild chart the rapidity lies strictly above the Newtonian value $r_s/(2r)$, and the fractional excess over that value is strictly less than $r_s/(r - r_s)$. A solar mass at one astronomical unit keeps the fraction below $1/(4 \times 10^7)$. The Milky-Way stand-in, taken at the compactification radius of Section 12.5, keeps it below 10^{-6}.

4.6 Rotor Action in the Dual Spacetime

The sole algebraic degrees of freedom considered here are the rotor angles within $\mathbb{H}\otimes_{\mathbb{R}}\mathbb{H} \cong \mathrm{Cl}(3,1)$ carried by every massive particle. Gravity, inertia, and quantum states are read as deviations and coherences among these angles.

The dual spacetime, related to the usual spacetime by the dual map $X \mapsto Xi$, inherits a time-reversed orientation. Consequently, an angle that generates a pure spatial rotation in the dual sector must be appropriately dual-transformed when incorporated into the unified rotor description.

Consider a dual-sector angle θ. In the pure dual basis, it produces a rotation generated by the elliptic versor $\hat{q}$ ($\hat{q}^2 = -1$):

$$\exp\left(\frac{\theta}{2}\hat{q}\right).$$

However, since the dual map reverses the arrow of time, the physical effect of this angle on the dual spacetime is equivalent to a boost when viewed through the dual transformation.

To reflect this correctly in the torsional mismatch calculation, we apply the dual conversion to the angle itself: $\theta \mapsto \theta i$. This transforms the dual contribution into a hyperbolic boost term:

$$\exp\left(\frac{\theta i}{2}\hat{q}\right) = \exp\left(\frac{\theta}{2}\hat{q}i\right) = \exp\left(\frac{\theta}{2}(q_1 iI + q_2 iJ + q_3 iK)\right).$$

Thus, the dual rotor is defined as

$$R_{\text{dual}} = \exp\left(\frac{\theta}{2}\hat{q}\right),$$

while its action on dual spacetime vectors, after the angle dual conversion $\theta \mapsto \theta i$, manifests as a boost generated by $i\Gamma_a$ ($(\hat{q}i)^2 = +1$).

In free fall, $\Omega = 1$ requires $R_{\text{dual}} = R_{\text{usual}}$, not $J = 0$. Mass-energy is the cost of deviations from that condition, and $J > 0$ is attractive gravity.

This angle-centric treatment, with explicit dual conversion $\theta \mapsto \theta i$, preserves the original form of the total rotor

$$R_{\text{total}} = \exp\left(\frac{\phi}{2}\hat{p} + \frac{\theta}{2}\hat{q}\right)$$

while correctly incorporating the time-reversal duality: the dual rotor R_{dual} generates rotations in its native basis, but its physical effect on the dual spacetime—through the dual-transformed angles—is that of a boost, naturally opposing the usual-sector boosts in vacuum.

4.7 Physical Interpretation and Gravitational Engineering

Gravity is the energetic cost of dual-rotor misalignment. Because J depends solely on the relative rotor Ω, dual-only modulation of θ cannot raise J. It conserves $J+M$, and dual-only shielding is possible precisely when the dual wall is nonpositive; a strictly negative wall admits repulsion. Gravitational neutrality is moreover bounded in mass: no admissible configuration with $J = 0$ carries more than five sixths of the cone's mass ceiling, whereas full repulsion reaches that ceiling unimpeded. The exact (J, M) image of the cone is the three-lobed domain of Section 13. On a boosted axis the clock factor γ_{eff} falls with dual excitation and meets 1 before the coaxial shield; a cross-axis shield cancels J while leaving that clock at $\cosh\alpha$. Whether an external field supplies that dual-channel force is a conjecture, taken up in that section. The continuum remains a macroscopic description; spacetime degrees of freedom in this theory are particle-local and paired.

5 Unified Description of Inertia and Gravity

In this section, we reveal the deepest consequence of the dual spacetime formalism: inertia and gravity are not merely equivalent — they are identical phenomena arising from the torsional mismatch between the usual and dual spacetimes. The distinction between inertial forces in accelerated frames and gravitational forces in curved spacetime dissolves completely. Both are the same dual rotor misalignment, with the equivalence principle emerging as a direct algebraic necessity rather than a postulate.

5.1 Inertial Resistance as Dual Rotor Rigidity

What the particle action calls rest mass is a stiffness of the lag between the two rotors. It is not the unsigned torsional mass M, and it is not the Killing scalar J. Free fall remains the coincidence of the rotors, $\Omega = 1$.

On the written action both rotors receive the same angular acceleration. The mass does not open a lag. A lag already present accelerates the sum of the rapidities in proportion to m, and while the lag itself does not drift the potential stands still. On one axis that potential is fixed by the unsigned mass and the square of the common rapidity (19), and it is not the mismatch J. Across axes the separation is

sharper. A usual rapidity on one axis, cancelled by an equal dual rapidity on another, has $J = 0$, positive unsigned mass, and a positive potential. Gravitational neutrality does not extinguish the stiffness, and it is not free fall.

The model in which the stiffness restores the lag takes both kinetic signs positive. There the mismatch oscillates and the common rapidity moves uniformly. That is not the written action, and neither reading is Newton's second law. Inertia, in this theory, is not a property of a continuum reacting to a mass. It is the way a lag already carried by the particle's own paired rotors accelerates their common rapidity.

5.2 Massless particles

Vanishing stiffness removes the potential and, with it, the acceleration of the common rapidity. Both angular accelerations are zero whatever the angles are. The rotors are not thereby forced to coincide, and neither J nor M is forced to vanish: a pure boost still carries a nonzero mismatch, a nonzero unsigned mass, and a relative rotor distinct from the identity. A photon, read as a particle of vanishing stiffness, pays no such acceleration. Resonance, the vanishing of J, and free fall are further conditions. The constancy of the speed of light is the statement that a change of frame costs no stiffness. It is not the statement that every massless configuration is the vacuum.

5.3 The Equivalence Principle as Algebraic Identity

Consider Einstein's elevator accelerating upward with acceleration a. An observer inside feels a downward force. Whether the floor pushes the usual rotor, or a neighbouring mass holds it off free fall, the weight that is felt is one and the same mismatch of the paired rotors.

In a gravitational field, with the elevator at rest on Earth, the floor must accelerate upward relative to free fall in order to remain stationary, and the mismatch of the paired rotors is the mismatch of uniform acceleration in flat spacetime.

Thus the equivalence principle is not an empirical coincidence but an algebraic identity: gravitational and inertial forces are indistinguishable because they are the same dual torsional phenomenon. Free fall is the coincidence $\Omega = 1$. It is not the vanishing of J. A particle is not in free fall merely because its stiffness vanishes, and a massive particle is in free fall when the two rotors agree.

5.4 Consequences and Predictions

- The arrow of time distinction between usual and dual spacetimes explains why massive particles have a preferred rest frame: the dual time reversal breaks perfect symmetry, generating the mass gap.
- Synchronising the dual rotor θ with ϕ can cancel the mismatch J. Dual-only, that cancellation is purchased one-for-one by unsigned mass M: it is not a reduction of M, and the dual budget may be parked on a silent axis. Reducing both J and M requires unwinding the two rotors together, toward the origin. The cancellation is also capped: a configuration with $J = 0$ cannot carry more than five sixths of the maximal unsigned mass, so a sufficiently heavy body must be made lighter before its inertial signature can be cancelled at all. Either operation is a kinematic possibility on the cone, not yet a laboratory coupling—a direct pathway, if a dual-channel force exists, toward propellantless control of inertia.
- The theory predicts that sufficiently high-frequency fields coupling to the dual spacetime could make massive objects behave as effectively massless over short timescales, allowing transient superluminal phase velocities (without violating causality, as information remains luminal).

Inertia and gravity are unified not by geometry, but by the internal torsional dynamics of particle-intrinsic dual spacetimes. The stiffness m and the unsigned mass M remain distinct quantities. With that distinction, both inertia and gravity cease to be fundamental — they become engineerable.

The dream of gravitational and inertial control is no longer science fiction; it is a direct consequence of abandoning the continuum and recognizing spacetime as paired, particle-local, and torsionally active.

5.5 Upper Limit of Acceleration and the Inertial Reversal Phenomenon

The double spacetime theory predicts, as one of its central falsifiable consequences, that acceleration itself possesses an absolute upper limit. The usual-sector boosts are non-compact ($\phi \in \mathbb{R}$) and, taken alone,

would permit unbounded rapidity. The dual sector, however, is intrinsically compact: the dual generators Γ_a square to -1, so the dual rotor $R_{\text{dual}} = \exp(\theta/2\,\hat{q})$ is governed by the elliptic phase $\theta \bmod 4\pi$. The written stiffness does not make θ track ϕ. The two rotors receive the same angular acceleration, and the lag is free, so a ceiling on proper acceleration is not a consequence of that tracking. We denote by a_{crit} a material-intrinsic ceiling, if one exists. Unlike the speed of light it would not be universal: it would depend on the particle and on the compactification of its dual sector. The written action does not supply it, and it does not yield $F = ma$.

What remains a proposal is a topological wrap of the compact dual phase. At the instant a sustained acceleration drives θ through the boundary at 4π, the dual rotor re-sutures its phase on the opposite side of the S^3-like dual sector. The proposal is that this wrap inverts the sign of the torsional mismatch bivector Ω_{biv}, and therefore reverses the direction of the inertial response over a single dual cycle. We refer to this as the inertial reversal phenomenon: a sudden flip of the response, triggered if the applied acceleration crosses a_{crit}. The torsional strain stored just below threshold would be released abruptly during the wrap, as an impulsive burst of internal energy in addition to the kinematic reversal itself.

We propose that this mechanism is observationally realised in the anomalous airbursts of large meteoroids. The hypersonic deceleration experienced by bolides entering the Earth's atmosphere can reach values that, for the constituent atoms of the body, are plausibly of order a_{crit}. Long-standing puzzles in the bolide community—most prominently the Tunguska event (1908) and the Chelyabinsk superbolide (2013), whose disintegration altitudes and total radiated energies are notoriously difficult to reconcile with kinetic-energy dissipation models based on aerodynamic ablation and fragmentation alone—are naturally explained as collective inertial reversals: when the deceleration of the meteoroid crosses the material-specific a_{crit}, the dual phases of its constituent atoms wrap simultaneously, dumping the accumulated torsional strain energy as an explosive, predominantly radiative pulse. The dependence of a_{crit} on composition would also account for the observed variability in airburst altitudes and yields between different meteorite classes (e.g. carbonaceous versus ordinary chondrites). A quantitative derivation of a_{crit} from the particle-intrinsic compactification scale remains open.

6 Strong-Field Regime: Layered Gravitational Reversal and the New Black Hole Paradigm

The double spacetime theory predicts a radically new structure for the endpoints of stellar collapse. The Killing form's sign asymmetry — positive for usual-spacetime boost dominance (attraction) and negative for dual-spacetime rotation dominance (repulsion) — does not simply reverse gravity once and for all. Instead, as density increases inward, the torsional mismatch Ω_{biv} oscillates in dominance between the two sectors, producing alternating layers of attraction and repulsion. This yields a stratified, onion-like interior: attraction $\to$ repulsion $\to$ attraction $\to$ repulsion $\to$ ..., with exactly one reversal per period of the interference factor. The amplitude of that alternation is settled in closed form in Section 8.1: the barriers separating the layers sharpen without bound toward the centre, while the plateaux between them are exponentially screened, so the interior is not a region of ever-stronger force but a stack of ever-sharper walls enclosing ever-quieter interiors.

6.1 Layered Torsional Structure from the Killing Form

The scalar $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ changes sign each time the dominant contribution flips from $i\Gamma_a$ (boost-like, $+$) to Γ_a (rotation-like, $-$). At extreme densities, the dual rotor parameters θ grow nonlinearly, driving successive resonances that flip the sign of J repeatedly as radius decreases.

The resulting effective potential features:

- Outer layers: standard attractive gravity ($J > 0$),
- First reversal: repulsive shell ($J < 0$) that halts infall (core bounce in supernovae),
- Deeper layers: attractive zones that re-accelerate matter inward, separated from the repulsive shells by sharp barriers,
- Alternating deeper shells, with barrier strength growing and plateau strength decaying, culminating in chaotic torsional turbulence.

On the admissible cone $|J|$ cannot exceed $\frac{3\pi^2}{8}$. The classical radial rapidity $\varphi \to \infty$ as $r \to r_s^+$ is an internal coefficient, and it leaves the cone. The expressed mismatch is the admissible point closest to that coefficient, so the growth seen in the chart is not a growth of the expressed mismatch.

6.2 Quasi-Horizon Cutoff

Classical Schwarzschild has $\varphi \to \infty$ and $A \to 0$ as $r \to r_s^+$. Identifying the three-axis admissible ceiling $|J| = \frac{3\pi^2}{8}$ with a radial pure-boost slice $J = \frac{1}{2}\varphi^2$ inverts to the finite cutoff

$$\varphi_{\max} = \frac{\pi\sqrt{3}}{2}, \qquad A_{\min} = \mathrm{e}^{-2\varphi_{\max}} = \mathrm{e}^{-\pi\sqrt{3}}.$$

The corresponding radius is $r_\star = r_s/(1 - A_{\min})$, equivalently $r_\star/r_s = (1 - A_{\min})^{-1}$. One has $0 < A_{\min} < 10^{-2}$ and $1 < r_\star/r_s < 1.005$. The saturated chart freezes $\varphi = \varphi_{\max}$ and $A = A_{\min}$ for $r \le r_\star$, and uses the classical formulae for $r > r_\star$. At $r = r_\star$ the two descriptions agree. The saturated factor A remains strictly positive: a true one-way horizon $A = 0$ does not form. (By contrast the classical factor vanishes at $r = r_s$.)

This cutoff is a working identification of the three-axis ceiling on a single boost axis. A one-axis admissible boost saturates only at $\varphi = \pi/2$, where $J = \pi^2/8 = \frac{3\pi^2}{8}/3$. The inverted value $\varphi_{\max} > \pi/2$ therefore lies outside the admissible cone. The construction saturates the three-axis bound, not the single-axis cone.

On the vacuum chart $r^2 T = -2$ at every radius and for every mass, so the cutoff is not what keeps T finite. It bounds φ and holds $A \ge A_{\min} > 0$. Repulsive layers remain the negative sign of J, and $|J| \le \frac{3\pi^2}{8}$ on the cone, so the mismatch of a repulsive layer is bounded. That bound is not a window for T.

Inside $r \le r_\star$ the redshift floor $A = A_{\min} > 0$ is held fixed, so $A' = 0$ and the field seed J_{field} vanishes. Equation (2) then gives $T = -2A_{\min}/r^2$, which does not vanish: the constant-A coframe still carries the angular anholonomy of the spherical frame. A constant Cartesian boost is the contrasting case, with $J \neq 0$ and $T = 0$.

This layered structure prevents both singularities and event horizons. Matter never reaches a central singularity. Instead, the core settles into a quasi-stable, chaotic condensate of extreme-density torsional energy, continuously oscillating between attractive and repulsive phases.

Conventional Kerr-type rotating black holes are ruled out: the theory forbids global frame-dragging of the required magnitude, as angular momentum is redistributed into dual rotor longitudinal modes rather than spacetime rotation.

6.3 Reinterpretation of Core-Collapse Supernovae and Remnant Formation

In massive star collapse, the first repulsive layer ($J < 0$) triggers explosive bounce at nuclear densities, powering the supernova without fine-tuned neutrino physics. Deeper attractive layers then recapture portion of the ejecta, forming the stratified remnant. The observed diversity of supernova energies and remnant masses emerges naturally from the number and strength of torsional layers excited.

6.4 Pulsars as Longitudinal Dual-Torsion Oscillators

Observed neutron stars are not rapidly rotating magnetized conductors. The double spacetime theory denies both millisecond spin periods and dipolar magnetic fields as the primary energy source.

Instead, pulsar radio pulses arise from ultra-high-frequency longitudinal oscillations of the dual rotors θ on the star's surface. These torsional waves propagate radially at nearly c, modulated by the chaotic interior layering, producing beams via curvature emission in the torsional shock zones.

Key predictions that surpass the rotating magnetar model:

- The theoretical minimum pulse period is determined by the Planck-scale double rotor frequency, which is orders of magnitude shorter than the observed ~ 1 ms limit, allowing us to explain short-period pulsars that could not be explained by the rotating pulsar picture.
- Pulsar "glitches" and period derivatives $\dot{P} < 0$ (period shortening) are the result of the pulsar losing energy causing the vibrational interface to shrink, accelerating the vibrational frequency over time, a natural mechanism not seen in rotation-only models.

- Magnetar flares and fast radio bursts are violent flips between attractive/repulsive layers, releasing torsional strain without requiring 10^{15} G fields.
- The death line in the P-$\dot{P}$ diagram corresponds to the point where interior layering stabilizes and longitudinal modes damp below detectability.
- The neutron star maximum mass is capped not by quark matter but by the same torsional layering that limits nuclear stability, predicting a strict upper bound $\sim 2.5M_\odot$ and explaining the absence of pulsars above $\sim 3M_\odot$.

The Crab pulsar's observed spin-down luminosity is reinterpreted as torsional wave leakage rather than magnetic braking, perfectly matching the energetics without invoking unrealistic magnetic field decay.

6.5 The New Black Hole Paradigm: Torsional Stars

What observers call "black holes" are in fact torsion stars — quasi-stable, layered condensates with no event horizon and no central singularity. The innermost chaotic core is a Planck-density plasma in perpetual torsional turbulence, continuously radiating via rotor tunneling (modified Hawking process). Information is preserved in the dual rotor phases.

Gravitational-wave signals from mergers feature repeated echoes from internal layer reflections, with characteristic frequencies tied to the torsional oscillation spectrum — already hinted in LIGO/Virgo data reanalyses.

The theory thus resolves the information paradox, explains the absence of Kerr spin signatures, and predicts that the most massive "black holes" ($\gtrsim 100M_\odot$) will exhibit periodic modulations in accretion luminosity as inner repulsive layers periodically eject material.

6.6 Central Floating Core: Zero Time Dilation at the Center

A common point of confusion in discussions of stellar interiors is the behaviour of gravitational time dilation. In GR, for a spherically symmetric static mass distribution, gravitational time dilation increases monotonically toward the centre: proper time runs most slowly at $r = 0$, where the gravitational potential is deepest.

In the double spacetime theory, the situation is markedly different. Because there is no background continuous spacetime manifold, gravitational effects—including both the effective acceleration and time dilation—arise solely from the torsional mismatch between the particle-intrinsic usual and dual rotors.

At the exact geometric centre of a spherically symmetric torsion star, perfect symmetry forces the usual-spacetime boost parameters ϕ_a induced by mass elements in opposite directions to cancel completely. The residual mismatch angle $\delta\phi_a = \phi_a - \theta_a$ therefore vanishes at $r = 0$, yielding $J = 0$ and a complete restoration of torsional balance.

Consequently:

- The effective gravitational acceleration is zero at the center (in Newtonian mechanics, GR, and DST).
- Gravitational time dilation is also zero at the centre: proper time at $r = 0$ runs at the same rate as in the vacuum far from the star.

This vanishing of time dilation at the core is a distinctive, observable-in-principle prediction of the double spacetime theory. It contrasts sharply with GR, where the central region experiences the strongest time dilation.

The zero-dilation central core also contributes to the stability of torsion stars by eliminating the infinite central compression and extreme clock-rate gradients that lead to singularities in the classical GR description.

6.7 Implications for Planetary and Stellar Cores: Instability and Fluctuations

The floating core paradigm, where $J(r = 0) \approx 0$ and effective gravity vanishes at the center, implies that the cores of planets and stars are not inert, stable regions but dynamically active zones prone to torsional instability. Unlike the monotonic increase in gravitational potential assumed in continuum models, the double spacetime theory predicts that the central floating core is a metastable state, susceptible to

perturbations that trigger oscillatory transitions between attractive and repulsive torsional layers. These fluctuations arise from the delicate balance of dual rotor phases in the isotropic core, where even minor density variations or external influences can shift the resonance condition $\beta(0) \to 1$, leading to transient excursions of J away from zero. Such excursions manifest as localized bursts of torsional energy, effectively generating mini torsion stars within the core. These mini torsion stars can induce seismic activity, magnetic field fluctuations, and energy transport anomalies observable at the surface. In stellar interiors, these core fluctuations may contribute to phenomena such as solar flares, sunspots, and variability in luminosity, as the torsional energy propagates outward through the layered structure. In planetary contexts, core torsional instability could explain geomagnetic reversals, mantle plumes, and episodic volcanic activity, as the torsional dynamics couple to convective processes in the overlying layers. Thus, the floating core is not a quiescent region but a crucible of dynamic torsional phenomena with far-reaching implications for planetary and stellar behavior.

7 Time Dilation in Double Spacetime Theory: The Dynamical Role of Dual-Spacetime Proper Time

Time dilation in the double spacetime theory emerges from the interplay between the usual spacetime and the dual spacetime, each endowed with its own proper time. The usual spacetime proper time τ governs the familiar kinematics observed in macroscopic Minkowski space, while the dual spacetime proper time $\tilde{\tau}$, encoded in the compact rotor phases θ, drives the internal torsional dynamics that manifest as gravitational and inertial effects.

7.1 Derivation of the Effective Lorentz Factor

The effective Lorentz factor $\gamma_{\text{eff}} = dt/d\tau$ is derived directly from the algebraic action of the total rotor

$$R_{\text{total}} = R_{\text{usual}} R_{\text{dual}},$$

where

$$R_{\text{usual}} = \exp\left(\frac{\phi}{2}\hat{p}\right) = \cosh\frac{\phi}{2} + \hat{p}\,\sinh\frac{\phi}{2}, \quad \hat{p}^2 = +1,$$

generates boosts in the usual spacetime, and

$$R_{\text{dual}} = \exp\left(\frac{\theta}{2}\hat{q}\right) = \cos\frac{\theta}{2} + \hat{q}\sin\frac{\theta}{2}, \quad \hat{q}^2 = -1,$$

generates rotations in the dual spacetime.

To reveal the origin of the interference term explicitly, consider the pure time vector $X = ct\,j$ ($j^2 = -1$) in the usual spacetime basis. The transformed vector is $\tilde{X} = R_{\text{total}}\, X\, R_{\text{total}}^{\dagger}$.

Due to the time-reversal duality, the inverse dual map from dual to usual spacetime is $X \mapsto X(-i)$. This map naturally yields the alignment $\hat{q}(-i) = -\hat{p}$ in the vacuum limit, transforming the dual rotor into an effective opposite-oriented rotor

$$R_{\text{dual}}^{\text{eff}} = \cos\frac{\theta}{2} - \hat{p}\sin\frac{\theta}{2}.$$

The negative sign in the sin term arises directly from this inverse mapping, reflecting the anti-synchronization required for torsion-free flat spacetime.

The total effective rotor is then

$$R_{\text{eff}} = R_{\text{usual}} R_{\text{dual}}^{\text{eff}} = \left(\cosh\frac{\phi}{2} + \hat{p}\sinh\frac{\phi}{2}\right)\left(\cos\frac{\theta}{2} - \hat{p}\sin\frac{\theta}{2}\right).$$

Expanding gives the scalar part

$$s = \cosh\frac{\phi}{2}\cos\frac{\theta}{2} - \sinh\frac{\phi}{2}\sin\frac{\theta}{2}$$

and the bivector coefficient

$$b = -\cosh\frac{\phi}{2}\sin\frac{\theta}{2} + \sinh\frac{\phi}{2}\cos\frac{\theta}{2}.$$

The net scaling of the time component arises from the squared magnitude of R_{eff} acting on the time vector via the sandwich product. For a rotor $R_{\text{eff}} = s + b\hat{p}$ with $\hat{p}^2 = +1$ and $\hat{p}$ anticommuting with the time generator j, the sandwich action on $X = ct\, j$ yields a time component scaled by $s^2 + b^2$, defining the effective Lorentz factor unambiguously as

$$\gamma_{\text{eff}} \;\equiv\; s^2 + b^2.$$

It is convenient also to introduce the scalar interference factor

$$\gamma_s \;\equiv\; s \;=\; \cosh\frac{\phi}{2}\cos\frac{\theta}{2} - \sinh\frac{\phi}{2}\sin\frac{\theta}{2},$$

which, unlike γ_{eff}, may change sign and will serve as the order parameter distinguishing attractive ($\gamma_s > 0$) from repulsive ($\gamma_s < 0$) torsional regimes below. The negative sign in the interference term is fixed by the time-reversed duality of the dual spacetime.

A short calculation collapses the squares:

$$\gamma_{\text{eff}}(\alpha, \beta) = \cosh\alpha - \sinh\alpha \sin\beta. \tag{4}$$

This factor is always at least $e^{-|\alpha|}$, hence never zero: a clock may be slowed or hastened, but it does not stop. If the dual sector is unexcited, $\beta = 0$, one recovers the special-relativistic factor $\cosh\alpha$. If there is no usual boost, $\alpha = 0$, then $\gamma_{\text{eff}} = 1$. The balanced locus $\alpha = \beta$, where $J = 0$ on one axis, is a different kinematics: $\gamma_{\text{eff}}(\alpha, \alpha) \neq \cosh\alpha$ as soon as $\alpha \neq 0$. Special relativity is the unexcited dual sector $\theta = 0$, not the vanishing of J.

7.2 Vacuum Limit and Massless Particles

Torsion-free kinematics is $\theta = 0$ (equivalently $R_{\text{dual}} = 1$ on a pure usual boost), which yields $\gamma_{\text{eff}} = \cosh\phi$. Balanced massive states have $J = 0$ with $\theta = \phi \neq 0$ and a different factor $\gamma_{\text{eff}} = \cosh\phi - \sinh\phi\sin\phi$. Photons, in this reading, maintain $\theta = 0$ at all velocities: they pay no dual-rotor price for a change of frame.

7.3 Massive Particles and Gravitational Effects

For massive particles the deviation $\delta\theta = \theta - \phi$ modulates γ_{eff}. We read that modulation as gravitational redshift. The stiffness of the action does not force the dual rotor to fall behind the usual one: on the written kinetic signs the two angular accelerations agree.

7.4 Sign Inversion of the Scalar Interference Factor and Repulsive Layers

When dual-spacetime rotations dominate ($\theta/2$ sufficiently large), the sinusoidal term can overpower the hyperbolic growth:

$$\tan\left(\frac{\theta}{2}\right) > \coth\left(\frac{\phi}{2}\right).$$

Then γ_s becomes negative, while $\gamma_{\text{eff}} = s^2 + b^2$ remains strictly positive. The sign of γ_s is the order parameter of repulsive layers. It is not a reversal of causality: proper time never runs backward.

This θ-centric formulation makes manifest that time dilation is fundamentally a torsional phenomenon originating in the particle-intrinsic dual spacetime. The usual rapidity ϕ provides the kinematic baseline, while the compact dual angle θ modulates that baseline. Those two displacements do not return together: the clock factor meets 1 at a dual angle strictly before the coaxial shield, and a cross-axis shield that cancels J leaves the boosted clock at its special-relativistic value. External coherent excitation of θ therefore acts on time dilation and on gravitational mismatch as two distinct kinematic coordinates.

8 Scale-Invariant Torsional Layers: The Strong Force as Gravitational Repulsion and Nuclei as Primordial Torsion Stars

The layered reversal is scale-invariant in the dimensionless variable $x = \ell/(2r)$. Under the equal-scale ansatz $\alpha(r) = \beta(r) = \ell/r$ the entire sequence depends on x alone; femtometer and MeV scales enter only through a characteristic length ℓ and the particle masses. The same mechanism that stratifies

a collapsing star is therefore available at nuclear densities. We propose that this is the strong force: gravity in its layered phase, not a second coupling. The Newtonian gravity fine-structure constant $\alpha_G = Gm_p^2/\hbar c \sim 10^{-38}$ must not be conflated with an $O(1)$ nuclear strength. Near a torsional node, $1/\gamma_s$ diverges; that is a kinematic amplification of a channel already present, not a derivation of the strong coupling from G.

One structural point must be settled at the outset. The equal-scale ansatz sets $\alpha_a = \beta_a$ on every axis, so J vanishes identically while M stays positive whenever some rapidity is nonzero. Nuclear matter, in this reading, sits on the balanced locus of Section 4: massive, with $\Omega \neq 1$, and with $J = 0$. What alternates across the layers is not J but the interference factor γ_s of Section 7. Attraction and repulsion below are the sign of γ_s. J and γ_s are independent order parameters; nuclear layering belongs to the second.

8.1 The Closed-Form Layer Spectrum

Written in the dimensionless variable $x = \ell/(2r)$, the equal-scale interference factor is

$$\gamma_s(x) = \cosh x \cos x - \sinh x \sin x.$$

Its derivative collapses to a single product,

$$\gamma_s'(x) = -2 \cosh x \, \sin x, \tag{5}$$

so the entire layer structure is governed by the zeros of sin. The critical points of γ_s are exactly the integer multiples of π, and the extremal values are

$$\gamma_s(n\pi) = (-1)^n \cosh(n\pi), \qquad n = 0, 1, 2, \ldots$$

On each interval $[n\pi, (n+1)\pi]$ the factor is strictly monotone—decreasing for even n, increasing for odd n—so each interval carries exactly one torsional node, and the sign of γ_s strictly alternates from one plateau to the next. Plateaux never vanish; only finitely many nodes lie in any bounded range of x.

Ordered outward in radius the pattern is: attraction for $r > r_1$; a barrier at r_1; repulsion on (r_2, r_1); a barrier at r_2; attraction on (r_3, r_2); and so on. The onion of Section 6 is not an extra assumption. It is this derivative, one barrier per π-interval of x, with no two consecutive layers of the same sign.

A node forces $\tan x = 1/\tanh x > 1$, so the n-th node sits in

$$n\pi + \frac{\pi}{4} < x_n < n\pi + \frac{\pi}{2},$$

or in radius

$$\frac{\ell}{(2n+1)\pi} < r_n < \frac{2\ell}{(4n+1)\pi}.$$

Thus $r_n = \Theta(1/n)$, every node obeys $r_n < 2\ell/\pi$, and the outermost node is the largest layer radius. The first two nodes satisfy

$$\frac{1}{6} < \frac{r_2}{r_1} < \frac{4}{5\pi} \approx 0.255.$$

The plateau amplitudes grow at least geometrically, $(\cosh \pi)^n \leq \cosh(n\pi)$ with $\cosh \pi > 11$, while never exceeding $\mathrm{e}^{n\pi}$. Successive layers amplify by an ever larger factor that stays strictly below $\mathrm{e}^{\pi} \approx 23.14$. Because γ_s sits in the denominator of the effective force $F = k/(\gamma_s r^2)$, the mid-layer amplitude at the n-th plateau $r = \ell/(2n\pi)$ is

$$|F| = \frac{k}{\ell^2} \cdot \frac{4\pi^2 n^2}{\cosh(n\pi)},$$

which decreases in n for $n \geq 1$. What grows inward without bound is the barrier structure—the nodes at which γ_s vanishes and $1/\gamma_s$ diverges. The derivative $-2 \cosh x \sin x$ does not vanish on $(0, \pi)$, so each zero is simple: the inward force has a simple pole, and the radial potential falls logarithmically, without a lower bound. The plateaux between barriers become exponentially quiet, successive factors falling like $\mathrm{e}^{-n\pi}$. In radius the screening is $1/\cosh\big(\ell/(2r)\big)$: the deep interior of a torsional condensate is quasi-free between the poles. The poles interrupt a smooth crossing. Reading that interruption as nuclear confinement remains a proposal.

A barrier is a force divergence, not a clock freeze: at a node $\gamma_{\text{eff}} = \gamma_b^2 > 0$ by (4). The ratios r_{n+1}/r_n depend only on the dimensionless roots x_n, so they are identical at nuclear and at stellar scales; ℓ fixes the absolute radii and cancels from every ratio. That is the content of the scale invariance in the title of this section, and the only sense in which nuclear and stellar layering are the same structure.

8.2 Torsional Layers at Nuclear Densities

As baryon density approaches nuclear saturation $\rho_0 \sim (2\text{–}3) \times 10^{14}\,\mathrm{g/cm^3}$ (equivalently $n_0 \sim 0.16\,\mathrm{fm}^{-3}$), the dual rotor parameters θ dominate the mismatch. Densities of order $10^{18}\,\mathrm{g/cm^3}$ are not ordinary nuclear interiors. The sign flips of γ_s are organised by one observational constant, the sign-reversal scale r_1: the separation at which a slow nucleon pair changes from attraction to repulsion. Scattering potentials are repulsive through $\frac{1}{2}\,\mathrm{fm}$ and attractive from 1 fm, so $\frac{1}{2}\,\mathrm{fm} < r_1 < 1\,\mathrm{fm}$. The channel length is the definition $\ell = 2x_1 r_1$, and it lies strictly between $14/15\,\mathrm{fm}$ and $19/10\,\mathrm{fm}$. Every node then lies at or inside r_1, hence below 1 fm. The second node lies strictly below $\frac{1}{2}\,\mathrm{fm}$, so the core scale lies in the repulsive layer. The rapidity wall is r_1 times a factor strictly between 224/189 and 190/157. A reversal at $\frac{3}{5}\,\mathrm{fm}$ places that wall inside 1 fm, and a reversal at $\frac{9}{10}\,\mathrm{fm}$ places it outside.

The layer sequence at nuclear densities then reads:

- Outermost barrier: the first node $\gamma_s = 0$, the observed sign reversal of nucleon–nucleon scattering, at r_1 with $\frac{1}{2}\,\mathrm{fm} < r_1 < 1\,\mathrm{fm}$.
- Repulsive shell inside it ($\gamma_s < 0$), extending inward to the second node, whose radius is between one sixth and one quarter of the first.
- Intermediate attractive plateau ($\gamma_s > 0$). The observed binding energy per nucleon is $\mathrm{BE}/A \sim 8\,\mathrm{MeV}$ near iron; an optical-model well depth of order 40–50 MeV is a distinct input.
- Deeper layers: strictly alternating attraction and repulsion, one barrier per π-interval of x, with exponentially quieter plateaux and a nearly free interior between the poles.

Pauli's exclusion principle is no longer taken as primitive. A nonzero dual spinor and its spin-reversed partner are orthogonal and span the dual spinor space, so one spatial pattern holds that pair and no third orthogonal dual spinor. Two dual spinors on the same ray carry one spin vector, and their antisymmetrized product vanishes: they occupy only the symmetric state of total spin 1, not the singlet of total spin 0. The singlet is the pair of a ray with its reverse. That is the exclusion of a repeated dual spinor, not the spin-statistics theorem. Spatial overlap of the repeated ray remains a geometric reading: when two identical particles attempt it, the dual-rotor θ phases enter the first repulsive torsional layer and produce an effective fermi pressure. Bosons, lacking this phase rigidity, experience only the attractive layers until much higher densities.

In short: we propose that the strong force is gravity in its strong-field, layered phase. QCD color charge is then a misreading of torsional charge carried by dual-rotor phases; we do not identify α_s with the discrete lattice floor ε_N.

8.3 Atomic Nuclei as Primordial Torsion Stars

Every atomic nucleus is a microscopic analog of the torsion stars described in Section 6, and by the previous subsection the two share not merely a qualitative resemblance but the same dimensionless node sequence. Reading inward from the nuclear surface, the derived structure is:

- Outermost repulsive barrier at r_1: generates the hard-core repulsion in NN scattering and the nuclear surface tension,
- Repulsive shell just inside it, terminating at $r_2 < r_1/4$,
- Attractive plateau between r_3 and r_2: mediates binding between nucleons,
- Alternating deeper barriers and plateaux, the barriers ever sharper and the plateaux exponentially quieter, so that the innermost region is nearly interaction-free between confining walls.

The amplification is entirely in the barriers; the intervening interior is screened by $1/\cosh\big(\ell/(2r)\big)$.

Saturation of nuclear density is the one nuclear regularity the layer picture claims to explain, and it is also where the standard empirical constants disagree. Constant nucleon number density is exactly $R = r_0 A^{1/3}$: a uniform sphere of that radius has density $3/(4\pi r_0^3)$ for every A. Numerically $r_0 = 1.2\,\mathrm{fm}$ gives

$$0.13\,\mathrm{fm}^{-3} < \frac{3}{4\pi r_0^3} < 0.14\,\mathrm{fm}^{-3},$$

which is not the empirical saturation value $n_0 = 0.16\,\mathrm{fm}^{-3}$. The coefficient consistent with n_0 is $r_0^\star \approx 1.14\,\mathrm{fm}$. The two commonly quoted nuclear constants disagree at the ten-per-cent level; a layer identification must choose between them.

Charge independence, spin–orbit coupling, and magic numbers are read as resonance conditions in the dual-rotor spectrum. Discrete finiteness of the rotor image at fixed compactification N yields only $A_{\max}(N) < \infty$. The layer geometry itself yields no such cutoff: if $\ell \propto A^{1/3}$, the ratio of the nuclear surface to any layer radius is independent of A; if the reversal scale is A-independent, already at $A = 1$ the surface $r_0 = 1.2\,\mathrm{fm}$ lies outside every node, since every node lies at or inside r_1 and $r_1 < 1\,\mathrm{fm}$. The estimate $A \sim 300$ remains an external heuristic.

Nuclear fission and fusion are torsional layer transitions: fission occurs when an excited dual mode drives γ_s through a node from positive to negative, explosively repelling fragments; fusion succeeds when two nuclei tunnel through the outermost barrier into the attractive plateau behind it.

8.4 Experimental Signatures and Immediate Predictions

- Ultra-high-energy nucleus-nucleus collisions (LHC, RHIC) should produce transient $\gamma_s < 0$ fireballs that decay via explosive hadron jets rather than hydrodynamic flow — already hinted in anomalous flow data.
- Precision measurements of the nuclear equation of state above saturation density will reveal oscillatory behavior in pressure vs. density, with characteristic repulsive peaks at $\sim$ 2–4 ρ_0.
- The ratio of successive barrier radii is a pure number, confined to $(1/6,\, 4/(5\pi))$ for the first pair and independent of ℓ. The same ratio must therefore appear in the NN repulsive core and in the layer spacing of compact remnants; a measured mismatch between the two would falsify the scale-invariance claim without any appeal to ℓ.
- Because the plateaux are screened by $1/\cosh\big(\ell/(2r)\big)$, probes that resolve well inside the outermost barrier should see an interaction that weakens exponentially with decreasing distance, interrupted only at the discrete barrier radii.

The four interactions are not collapsed into a single force. They are two channels of one geometry. Gravity and electromagnetism share the layered reversal of γ_s and differ only in which phase of the dual rotor sources the mismatch: the kinematic pair (ϕ, θ) tied to inertial mass, or an independent pair (α, β) tied to charge. The strong force is the layered phase of the gravitational channel at nuclear densities; electron shells, in the next section, are the analogous layered phase of the electromagnetic channel. Faraday coefficients occupy the same six-generator space as the dual rotor (Section 11). A spatial axis splits that six-space into an $\mathfrak{so}(2,1)$ stabilizer and a complement that moves the axis; compact SU(2) in the six-space is the dual rotor, not a symmetry of a fixed chirality. A duality mix of the stabilizer's only compact generator with the parallel electric field preserves the axis only at a trivial angle.

There is only one geometry — the layered dance of attraction and repulsion in the dual rotor, played simultaneously in two phase channels. Atomic nuclei are the primordial torsion stars of the gravitational channel; atomic electron shells, as we shall see next, are the primordial torsion stars of the electromagnetic channel — and from this single dual-rotor geometry, expressed in two angular channels, all structure emerges. Each electron and each proton carries both channels, and exerts, at every separation, the gravitational force of its mass and the electric force of its charge.

9 Time Reversal and Electron Shells via Coulombic Torsional Layers

In the double spacetime framework, the Coulomb interaction between charged particles arises from torsional mismatch in the dual spacetime sector, as the gravitational interaction does. The two channels act together. The electric term has the form of the gravitational term, with the observational Coulomb constant $k = \alpha\hbar c/e^2$ and the charges standing where G and the masses stand. The gravitational term remains beside it. For a proton of charge $+e$ and an electron of charge $-e$, the electric term is

$$F(r) = \frac{ke^2}{\gamma_s(r)\, r^2},$$

where γ_s is the scalar interference factor introduced in Sec. 7, which retains the same algebraic structure from the dual rotor interplay (and which can change sign, unlike the proper-time Lorentz factor $\gamma_{\text{eff}} \geq 0$):

$$\gamma_s(\alpha(r), \beta(r)) = \cosh\left(\frac{\alpha(r)}{2}\right)\cos\left(\frac{\beta(r)}{2}\right) - \sinh\left(\frac{\alpha(r)}{2}\right)\sin\left(\frac{\beta(r)}{2}\right).$$

Here, $\alpha(r)$ and $\beta(r)$ are interpreted as electromagnetic rapidity and dual rotation angle parameters that scale inversely with distance: $\alpha(r) \propto 1/r$ and $\beta(r) = \lambda/r$, with λ a characteristic length scale set by the compact dual spacetime.

For the hydrogen atom ($Z = 1$), this yields

$$F(r) = \frac{ke^2}{r^2\left[\cosh\left(\frac{\alpha(r)}{2}\right)\cos\left(\frac{\beta(r)}{2}\right) - \sinh\left(\frac{\alpha(r)}{2}\right)\sin\left(\frac{\beta(r)}{2}\right)\right]}.$$

The formula just written is the electric term. The mass term is present at the same separation. With $\gamma_{s,G}$ the interference factor of the gravitational channel, read at the nuclear sign-reversal scale, the force that increases the separation of the electron and the proton is

$$F_r = -\frac{Gm_e m_p}{\gamma_{s,G}(r)\, r^2} - \frac{ke^2}{\gamma_s(r)\, r^2}.$$

Where the two factors have the same sign, the two terms point the same way, and F_r does not vanish. The Bohr radius lies on the outer plateau of both lengths, so both factors are positive there and both terms attract. The mass term is then smaller than the charge term by a factor strictly between 4.40×10^{-40} and 4.42×10^{-40}. The nodes below are the nodes of the charge channel.

One compactification integer, read as the electron's rest mass in Planck units, places the nonzero normalized floor at sixteen thirds of the gravitational fine-structure constant $Gm_e^2/\hbar c$. That constant lies strictly between 1.70×10^{-45} and 1.77×10^{-45}, and the electromagnetic constant exceeds the resulting floor by a factor strictly between 7.7×10^{41} and 8.1×10^{41}. Setting the same floor equal to the electromagnetic constant asks for an integer whose square lies strictly between 729 and 732, hence between 27 and 28. Setting the floor equal to $Gm_e^2/\hbar c$ asks for an integer whose square lies strictly between 3×10^{45} and 4×10^{45}. The algebra fixes neither integer. The scale this reading produces is the gravitational floor.

The hyperbolic-trigonometric form of γ_s introduces extreme oscillatory behavior. As r decreases:

- At large r, $\alpha(r), \beta(r) \to 0$, so $\gamma_s \to 1$, recovering the standard Coulomb attraction. The gravitational factor tends to 1 as well, and the mass term tends to the Newtonian attraction of the two masses.
- At smaller radii, the interference term generates points where $\gamma_s = 0$ (infinite repulsion) and regions where $\gamma_s < 0$ (force reversal to repulsion).
- These discrete radii r_n satisfy the resonance condition

$$\tanh\left(\frac{\alpha(r_n)}{2}\right)\tan\left(\frac{\beta(r_n)}{2}\right) = 1.$$

The force is undefined at a node. A node is a force divergence, not a clock freeze: $\gamma_{\text{eff}} = \gamma_b^2 > 0$ by (4). The same simple zero lets the radial potential fall without a lower bound, and the divergence does not by itself supply a circular orbit.

9.1 First Coulombic node and characteristic length

The electromagnetic length is the reduced Compton wavelength of the electron, an observational constant. Attraction at the Bohr radius places that radius in the outer well. The node is not that radius.

9.2 Coulombic circular orbits

The centripetal balance $mv^2/r = ke^2/(\gamma_s r^2)$ rearranges to

$$v^2 = \frac{ke^2}{m\,\gamma_s\, r}.$$

Circular speed is real only where $\gamma_s > 0$. On a repulsive layer the circular v^2 is negative. In the Coulomb limit $\gamma_s = 1$ one recovers $v^2 = ke^2/(mr)$. Bound circular orbits therefore exist only in attractive layers—the exterior Coulomb well $r > r_1$ and the inward attractive strata between later barriers.

Under the equal-scale ansatz the resonance condition is $\tanh x \cdot \tan x = 1$, the same as $\gamma_s = 0$, so the layer spectrum of Section 8.1 applies verbatim: radii scale as $r_n = \Theta(1/n)$, not as $\Theta(n^2)$. The first two nodes satisfy $r_2/r_1 < 1$, while Bohr bookkeeping $r_n = n^2 a_0$ gives $r_2/r_1 = 4$. The equal-scale tower runs inward; Bohr shells run outward. That order is the order of the rapidities, so the same inward placement holds for every common phase that decreases strictly with distance, and not only for λ/r. Matching the Rydberg series therefore requires a separate radius law such as $r_n = n^2 a_0$. The barriers are the geometry; the hydrogen spectrum is additional bookkeeping.

9.3 Z-contraction

On the equal-scale locus, $\ell \mapsto \ell/Z$ contracts every layer radius by Z, while $\ell \mapsto Z\ell$ expands them. The substitution $\alpha, \beta \propto Z/r$ at fixed ℓ is the expanding map: it evaluates the $Z = 1$ profile at r/Z and moves nodes outward. Observed shrinkage of electron shells with nuclear charge is the identification $\ell(Z) = \ell/Z$.

9.4 Energy Conservation in the Dual Structure

The apparent non-conservative behavior of the manifest potential $V_{\text{eff}}(r) = -\int F(r)\, dr$—with its violent oscillations and sign changes—is resolved by distinguishing between the manifest potential observed in the usual spacetime and the true potential stored in the compact dual-spacetime rotor phases θ_a.

Total energy is conserved across both sectors:

$$E_{\text{total}} = E_{\text{kinetic (usual)}} + V_{\text{eff}}(r) + V_{\text{true}}(\theta) = \text{constant}.$$

The true potential V_{true} is proportional to the torsional mismatch scalar $J \propto (\phi - \theta)^2$, acting as a hidden reservoir. During layer transitions, a decrease in V_{eff} (attractive to repulsive) is exactly compensated by an increase in V_{true} as the dual rotor winds, and vice versa.

Photon emission corresponds to the release of stored dual-phase energy when the rotor unwinds into a lower torsional state. This dual bookkeeping ensures strict energy conservation without violating the discrete, particle-local nature of spacetime. Observed spectral differences require the separate Bohr bookkeeping of shell radii discussed above; they are not implied by equal-scale $\gamma_s = 0$ alone.

The Coulombic force therefore accounts for discrete torsional barriers and for circular orbits in attractive layers, without a wave-function postulate. Identifying those barriers with atomic electron shells is a proposal: the layer spectrum is the geometry, its absolute scale and the Bohr bookkeeping are inputs.

10 Stability on the balanced locus

On a configuration with vanishing mismatch and positive unsigned mass the two rapidity budgets are equal, and each is the mass itself,

$$\sum_a \alpha_a^2 = \sum_a \beta_a^2 = M.$$

A motion that holds either budget fixed cannot reach the vacuum. Fixing the usual rapidities conserves $J + M$; fixing the dual angles conserves $M - J$. In either case a positive budget remains, and with it a positive mass. The configuration of vanishing mass differs from such a seed in both α and β.

The written one-axis flow has the same rest point. Vanishing mismatch, vanishing mismatch rate, and vanishing common velocity keep both angles fixed, so the unsigned mass does not move. A nonzero mismatch rate leaves that value at once and is not restored, because the written density frees δ. On the aligned locus the same flow keeps the mismatch at zero, and the unsigned mass is one quarter of the square of the common rapidity. That mass stands still if and only if the common rate vanishes. A nonzero common rate meets the vacuum once and does not stay there: except at that instant the two rotors still disagree. Alignment holds the mismatch. It does not hold the mass.

The hydrogen well does not pay this rest. Its ionisation energy is exactly $\alpha^2/2$ of the electron rest energy, and that fraction lies between two and three parts in one hundred thousand, so the rest energy is more than 3×10^4 hydrogen ionisations and fewer than 5×10^4. The exterior Coulomb layer can bind the electron. It cannot account for the electron.

Nor does the first node. The substitution $x = \lambda/(2r)$ turns the inward change of the radial potential into $-2ke^2/\lambda$ times $\int dx/\gamma_s$, and that integral is unbounded as x approaches the node. A circular orbit in the attractive well still has no continuation through it: on the repulsive side the written v^2 is negative, and at the node the force is undefined. What holds the electron is the equality of the two budgets, which one channel cannot spend, together with the smallness of this binding against the rest energy.

11 Electromagnetism and the Weak Channel in Dual Spacetime

Electromagnetism and the weak channel occupy the same six-generator space as the dual rotor. They are not the same map. The dual map $X \mapsto Xi$ is Hodge duality on that space: it exchanges the usual (electric, hyperbolic) and dual (magnetic, cyclic) summands, has period 4, and sends $J \mapsto -J$. Chirality, by contrast, requires a generator that squares to $+1$. The $\mathrm{Cl}(3,1)$ pseudoscalar satisfies $i^2 = -1$, so $(1 \pm i)/2$ are not idempotent and cannot serve as left/right projectors. The complementary pair built from the spatial axis e_1 with $e_1^2 = +1$,

$$P_L = \frac{1 - e_1}{2}, \qquad P_R = \frac{1 + e_1}{2},$$

is idempotent, sums to the unit, and multiplies to zero. Duality sends e_1 to the trivector $-e_0e_2e_3$, which is not a real multiple of e_1: the chirality axis is not a dual-invariant line. The product of the commuting square-$+1$ pair e_1 and e_0e_2 generates a left ideal of real dimension eight. That ideal is the direct sum of the four-dimensional Minkowski core $\mathrm{Cl}(3,1)\,P$ and a copy along the null generator e_4, hence reducible as a $G(3,1,1)$-module; we do not identify it with an irreducible Dirac module, nor claim $\mathrm{Cl}(3,1)$ irreducibility of the core.

The Faraday coefficients (E_a, B_a) occupy the same six-generator space as the dual-rotor bivector Ω_{biv}:

$$F = F_{\mathrm{usual}} + F_{\mathrm{dual}}, \qquad F_{\mathrm{usual}} = \sum_a E_a\, i\Gamma_a, \qquad F_{\mathrm{dual}} = \sum_a B_a\, \Gamma_a. \tag{6}$$

Electric field is usual-sector, boost-like, noncompact. Magnetic field is dual-sector, rotation-like, compact. Duality $X \mapsto Xi$ acts on coefficients by $(E, B) \mapsto (B, -E)$. On this identification the mismatch scalar is the Faraday quadratic

$$J(F) = \tfrac{1}{2}(E^2 - B^2), \qquad M(F) = \tfrac{1}{2}(E^2 + B^2).$$

A pure electric configuration has $J \geq 0$; a pure magnetic configuration has $J \leq 0$. That is the Killing-form signature of the two sectors, written in laboratory fields. Fields with $E^2 = B^2$ sit on the balanced massive locus $J = 0$, $M > 0$, not on the vacuum $M = 0$.

A circularly polarised wave of helicity $\sigma = \pm 1$ (units $c = 1$),

$$\mathbf{E} = E_0(\hat{x}\cos\psi + \sigma\hat{y}\sin\psi), \qquad \mathbf{B} = E_0(-\hat{x}\sin\psi + \sigma\hat{y}\cos\psi),$$

is null at every phase $\psi = kz - \omega t$: $J = 0$, $E \cdot B = 0$, and $M = E_0^2$. Its signed Poynting component along the beam is σE_0^2 at every phase. Laboratory time reversal $(E, B) \mapsto (E, -B)$ reverses that sign; duality preserves it. The two local Faraday invariants J and $E \cdot B$ vanish identically on the wave, so a helicity-odd scalar cannot be a polynomial in those invariants. Among the six classical Faraday quadratics $\{E^2, B^2, E \cdot B, P_x, P_y, P_z\}$, a combination odd under helicity reversal equals a multiple of P_z on every circular wave. Superposing the wave onto a background torsion shifts J by a first harmonic in the phase; the four-phase mean of J is the background value. Linear superposition does not displace mean mismatch.

Three maps act on this six-space and must not be conflated. Duality has period 4 and sends $J \mapsto -J$. The usual–dual parameter swap has period 2, also sends $J \mapsto -J$, and coincides with duality only on a pure magnetic field. Laboratory time reversal preserves J and flips $E \cdot B$. Same-axis electric and magnetic generators commute, so a simultaneous rotation of the pair (E, B) is well-defined:

$$E_a(\omega) = E_a\cos\omega + B_a\sin\omega, \qquad B_a(\omega) = -E_a\sin\omega + B_a\cos\omega,$$

with $J(\omega) = J\cos 2\omega + (E \cdot B)\sin 2\omega$. Duality is the special angle $\omega = \pi/2$.

Relative to a spatial axis e_a the six generators split into a commuting triple (B_a, E_{a+1}, E_{a+2}) and an anticommuting triple (E_a, B_{a+1}, B_{a+2}). The sandwich $P_R X P_R$ keeps the former and kills the latter; the selection is axis-dependent, not a Lorentz-scalar Weyl projection. The commuting triple is a Lie algebra: its brackets close, with the parallel magnetic generator compact and the two perpendicular electrics noncompact, so the algebra is $\mathfrak{so}(2,1)$ rather than $\mathfrak{su}(2)$. The anticommuting complement does not close;

the bracket of its first two generators is the remaining Cartan electric. Duality exchanges the two triples, up to signs. Among the six written generators the only triple in which every element squares to -1 is the dual-rotor (magnetic) triple itself, and those three satisfy the quaternion table $IJ = K$. Compact SU(2) in this six-space is therefore dual-sector rotation, not a chirality-selected Faraday Cartan. Three spatial axes each furnish such a split, and the associated projectors are complementary and idempotent, but projectors of distinct axes do not commute, nor do the three right projectors sum to the unit: they are alternative embeddings of the same $3 + 3$ geometry, not three simultaneous generation labels. The time axis squares to -1 and cannot serve as a chirality generator, for the same algebraic reason as the pseudoscalar. A circular wave is the phase quadrature of those two triples, so neither projector selects the helicity sign $\sigma = \pm 1$. Axis chirality is not photon helicity.

The commuting triple is the stabilizer of the axis. Rotation about the axis, and a boost perpendicular to it, leave e_a fixed. A dual rotation by an angle θ about the perpendicular generator B_{a+1} turns the axis in a spatial plane,

$$e_a \mapsto \cos\theta\, e_a - \sin\theta\, e_{a+2},$$

and the rotation about B_{a+2} turns it toward the remaining direction,

$$e_a \mapsto \cos\theta\, e_a + \sin\theta\, e_{a+1}.$$

At $\theta = \pi$ the axis has reversed, and P_L is exchanged with P_R. Of the three compact generators only B_a, the rotation about the axis itself, holds the axis still. The dual rotor moves handedness. It is not a gauge symmetry of a fixed projector.

A boost of rapidity α along the axis does not turn the axis into another spatial direction. It tilts the generator into time,

$$e_a \mapsto \cosh\alpha\, e_a + \sinh\alpha\, e_0,$$

and the image still squares to $+1$. The parallel electric generator is not itself a rotor: multiplied by its reverse it equals -1, while each magnetic generator is a rotor. What a fixed spatial axis selects inside this six-space is therefore the noncompact stabilizer $\mathfrak{so}(2,1)$, and the only compact direction in that stabilizer is the single generator B_a.

In the plane spanned by that generator and the parallel electric field, an element $c_B B_a + c_E E_a$ commutes with e_a if and only if $c_E = 0$. A duality mix of the pair leaves the stabilizer at every nontrivial angle. There is no chirality-preserving Weinberg angle in this plane. The stabilizer contains no second compact generator with which B_a could mix while holding the axis fixed.

The first-order sandwich increment of a rotor $R = 1 + \varepsilon\Omega$ is the commutator $\Omega X - X\Omega$, not the outer product $F \wedge X$. On a rest particle a pure electric generator produces a spatial kick and every magnetic generator produces none. On $X = e_0 + \mathbf{v}$ the commutator is the 4-vector

$$FX - XF = (E \cdot v)\, e_0 + (E - v \times B). \tag{7}$$

That is the Lorentz 4-force of the laboratory time-reversed field $(E, -B)$. The familiar 3-force $q(E+v\times B)$ differs from the sandwich by the sign of B; laboratory time reversal converts one into the other. A rest frame receives the electric kick E with vanishing power. A velocity along a circular beam yields a transverse kick with no longitudinal component. Maxwell's equations are not derived: the identity is kinematic on the sandwich.

The electromagnetic and weak channels therefore live in one six-space. The dual map exchanges the two sectors and is not a chirality projector. The sign of J distinguishes the noncompact electric sector from the compact magnetic sector. The sandwich on a particle's 4-velocity is the Lorentz force of $(E, -B)$. A mix ω rotating (E, B) on each axis is the duality circle, and at $\omega = \pi/2$ it swaps the Cartan $\mathfrak{so}(2,1)$ with the complement that moves the axis. It does not rotate two compact SU(2) factors into each other, and in the parallel plane it preserves a chirality axis only at a trivial angle. A dual-only increment of the dual angles occupies that six-space: superposing a magnetic Faraday field, and substituting $\theta_a \mapsto \theta_a + eA_a$ in the dual rotor, are two laboratory readings of it. Their kinematics occupy Section 13. An extended rotor built from $\int F\, ds$ remains open. The harmonic stiffness does not furnish a Higgs reading of rotor rigidity; that fact belongs to the particle action of Section 16. Neither the dual map nor the dual rotor supplies a V–A coupling.

12 Dark Matter-Free Galactic Rotation Curves from Dual-Spacetime Compactness

The same dual-rotor geometry that layers nuclei and electron shells unfolds, at cosmological scales, as compactness. The dual rotor is a three-sphere. We read the gravitational channel, driven by (ϕ, θ) and tied to inertial mass, as organizing that sphere into galaxy-scale patches of compactification radius R; the electromagnetic channel, driven by (α, β) and tied to charge, unfolds over substantially larger, near-cosmic scales. A spherical source on the three-sphere carries a radially harmonic cotangent potential, normalized by Gauss's law. Its acceleration is a sum over images of the source repeated with period πR. If those images are laid along the usual radius, the circular speed has no finite value at the antipode $r = \pi R$, which for a Milky-Way mass lies within the measured rotation curve. That placement makes the usual radius periodic, while compactness belongs to the dual spacetime alone. Laid instead along the dual direction, the same images give the acceleration $(GM/(rR)) \coth(r/R)$. If the compactification radius scales as $R = \sqrt{GM/a_0}$, this is a single relation between the observed and the Newtonian acceleration, and the rotation curve is the baryonic Newtonian curve multiplied by $x \coth x$, with $x^2 = a_0 r / v_{\mathrm{bar}}^2$. It rises from the centre, turns over, and flattens without a further rise, at a speed that obeys $v^4 = GMa_0$ exactly. If galaxies are such patches, the missing acceleration is this geometry, read from baryons alone. That identification is the proposed reading. The acceleration a_0 below is an external input, not a consequence of the dual-rotor algebra.

12.1 The compact dual spacetime

The dual rotor $R_{\mathrm{dual}} \in \mathrm{SU}(2) \cong S^3$ parameterizes the dual degrees of freedom intrinsic to every massive particle. Its three rotation angles are coordinates on a three-sphere of radius R, while the usual spacetime remains flat. The logarithm

$$\log : S^3 \to \mathfrak{su}(2) \cong \mathbb{R}^3$$

sends each rotor to a point of a Euclidean ball of radius πR. It is the three-dimensional azimuthal equidistant projection: faithful at the origin, and singular at the antipode, which is spread along the boundary of the ball. The volume distortion is the Jacobian

$$\mathcal{J}(\chi) = \frac{\sin^2 \chi}{\chi^2}, \qquad \chi = r/R.$$

As $\chi \to 0^+$, $\mathcal{J}$ tends to 1. At the antipode, $\mathcal{J}$ vanishes. A Newtonian calculation assumes $\mathcal{J} = 1$ at every radius. On a galactic patch that assumption holds only near the centre.

12.2 The cotangent potential

In geodesic polar coordinates on the three-sphere,

$$ds^2 = dr^2 + R^2 \sin^2(r/R)\,(d\vartheta^2 + \sin^2\vartheta\, d\varphi^2),$$

the area of a geodesic sphere is $4\pi R^2 \sin^2(r/R)$. The cotangent

$$\Phi(r) = -\frac{GM}{R} \cot\Big(\frac{r}{R}\Big)$$

is the radial potential whose flux through every such sphere equals $-4\pi GM$. Away from the origin the product of the area factor with $\partial_r \Phi$ is the constant GM, and the radial Laplacian therefore annihilates Φ:

$$\partial_r^2 \Phi + \frac{2}{R} \cot\Big(\frac{r}{R}\Big)\, \partial_r \Phi = 0.$$

Differentiating gives the acceleration

$$g(r) = -\frac{d\Phi}{dr} = -\frac{GM}{R^2 \sin^2(r/R)}.$$

On the open interval $0 < r < \pi R$ this acceleration is strictly negative. The force points toward the mass on both sides of the equator. The potential itself is negative if and only if $r < \pi R/2$, and it vanishes only at the equator. The equator is the edge of the negative well. It is not a reversal of the force. Past it, Φ

is positive and still slopes down toward the source; as the geodesic spheres shrink toward the antipode, the same flux is squeezed into a smaller area and $|g|$ grows, still inward.

For $0 < r \le R$ the germ at the origin is Newtonian. The potential differs from $-GM/r + GMr/(3R^2)$ by a remainder of order r^3/R^4. The correction $GMr/(3R^2)$ is the curvature of the three-sphere. It disappears in the decompactification $R \to \infty$, and with it the whole departure from the Newtonian potential.

12.3 Images along the radius

The acceleration is a lattice sum. The partial-fraction expansion of the cotangent gives

$$\frac{GM}{R^2 \sin^2(r/R)} = \sum_{n \in \mathbb{Z}} \frac{GM}{(r - n\pi R)^2}.$$

The term $n = 0$ is Newton's; the others are the source seen again at every multiple of πR along the radius. A circular orbit in the flat chart has

$$v^2(r) = r\,|g(r)| = \frac{GM}{R}\, f(r/R), \qquad f(x) = \frac{x}{\sin^2 x}. \tag{8}$$

On $(0, \pi/2)$ the shape has a single shallow minimum, at the root $x_0 \in (1.14, 1.20)$ of $\tan x = 2x$, with $1.35 < f(x_0) < 1.41$. Past it f rises, and as $x \to \pi^-$ it tends to $+\infty$: the image $n = 1$ reaches the orbit at the antipode. For a Milky-Way mass, under the scaling of Section 12.5, the antipode lies between 25 and 29 kpc, and measured rotation curves stay flat across such radii. The radial lattice makes the usual radius itself periodic, whereas only the dual spacetime is compact. It is the chart's reading of the images, faithful near the centre, and it is not the galactic law.

12.4 Images along the dual direction

The duality $X \mapsto Xi$ relates the elliptic geometry of the dual sector to the hyperbolic geometry of the usual sector by analytic continuation (Section 4); on the cotangent it acts as $i \cot(ix) = \coth x$. Carried through that continuation, the period πR leaves the usual radius and runs along the dual direction. Image n sits at squared distance $r^2 + (n\pi R)^2$, one leg in usual space and one around the compact dual circle, and the usual radius stays non-compact. The same cotangent expansion, at imaginary argument, sums this lattice in closed form:

$$g(r) = \sum_{n \in \mathbb{Z}} \frac{GM}{r^2 + (n\pi R)^2} = \frac{GM}{rR} \coth \frac{r}{R}. \tag{9}$$

The ratio to Newton is the dual enhancement

$$\eta(x) = x \coth x, \qquad x = r/R,$$

and for every $x > 0$

$$\max(1, x) < \eta(x) < \min\left(1 + x,\ 1 + \frac{x^2}{3}\right).$$

The enhancement increases strictly and tends to 1 at the origin, while coth decreases strictly to 1. Near the centre the two lattices agree: both begin as $GM/r^2 + GM/(3R^2)$, the curvature correction of the cotangent germ. The dual sum stays strictly below that germ at every radius, and strictly above both GM/r^2 and $GM/(rR)$. At large radius the second dominates. The acceleration falls as $1/r$ rather than $1/r^2$, and no radius is singular.

At a fixed compactification radius the lattice is linear in the source mass: the acceleration of a sum of masses is the sum of the accelerations, and scaling the mass scales the acceleration by the same factor. The fractional excess over the inverse square depends only on r/R. It is positive, and strictly less than $(r/R)^2/3$. On $(0, \pi)$ the three-sphere factor $(x/\sin x)^2$, the reciprocal of the volume Jacobian, lies strictly above the shared germ $1 + x^2/3$, while the dual enhancement lies strictly below it. A solar mass, on the same external inputs, has compactification radius between 7020 and 7040 astronomical units, so at one astronomical unit the dual fractional excess lies below 10^{-8}. At the Milky-Way compactification radius the exterior rapidity exceeds its Newtonian value by a fraction below 10^{-6}, beside a dual excess at $r = R$ between 0.31 and 0.32.

12.5 Baryonic Tully–Fisher and the acceleration relation

Let the compactification radius scale with the mass as $R = \sqrt{GM/a_0}$, where $a_0 \approx 1.2 \times 10^{-10}$ m/s^2 is the observed baryonic Tully–Fisher acceleration, taken as an input rather than derived from the dual-rotor algebra. A heavier galaxy is a larger patch. A derivation of a_0 is not claimed. About a point mass the circular speed is $v^2 = r\,g = (GM/R)\coth(r/R)$. It decreases strictly with radius toward

$$v_\infty^2 = \frac{GM}{R}, \qquad v_\infty^4 = GM\,a_0,$$

exactly: slope four, with no shape factor. The approach is from above and fast. Past $r = 3R$, v^2 lies within 1/200 of v_∞^2, so v is within a quarter of a percent of v_∞. There is no second rise and no antipode.

The same scaling turns the lattice into a radial acceleration relation. With $g_\mathrm{N} = GM/r^2$ one has $x^2 g_\mathrm{N} = a_0$ and $GM/(rR) = \sqrt{a_0 g_\mathrm{N}}$, so the acceleration is a function of g_N alone,

$$g = \sqrt{a_0\,g_\mathrm{N}}\,\coth\sqrt{\frac{a_0}{g_\mathrm{N}}}, \tag{10}$$

bounded by

$$\max(g_\mathrm{N}, \sqrt{a_0 g_\mathrm{N}}) < g < g_\mathrm{N} + \min\Big(\sqrt{a_0 g_\mathrm{N}},\ \frac{a_0}{3}\Big).$$

At high acceleration the excess over Newton stays below $a_0/3$. Tying the patch radius to the enclosed mass makes the relation strictly subhomogeneous and strictly subadditive: for $k > 1$, scaling g_N by k multiplies g by a factor strictly smaller than k, and the acceleration of two Newtonian contributions taken together is strictly smaller than the sum of the accelerations. While g_N is large compared with a_0, the inverse-square reading carries the acceleration, and the fractional excess becomes of order one as g_N passes through a_0. At low acceleration g approaches $\sqrt{a_0 g_\mathrm{N}}$, the deep regime of Milgrom's phenomenology. In between, the relation is fixed, with no interpolating function to choose. At $g_\mathrm{N} = a_0$, where $x = 1$, the enhancement is $\coth 1$, between 1.31 and 1.32.

For the Milky-Way stand-in $M \approx 6 \times 10^{10}\,M_\odot$ the same inputs give $8\,\mathrm{kpc} < R < 9\,\mathrm{kpc}$ and

$$175\,\mathrm{km/s} < v_\infty < 176\,\mathrm{km/s}.$$

To identify a_0 with a characteristic acceleration of the patch is still a proposal.

12.6 The rotation curve

At radius r the flux is that of the enclosed mass $M(r)$, and the patch that carries it has $R(r)^2 = GM(r)/a_0$. Outside the baryons this is the single patch of the total baryonic mass M_b; inside, it shrinks with the enclosed mass. With $g_\mathrm{N} = v_\mathrm{bar}^2/r$, the orbit sits at $x = r/R(r) = \sqrt{a_0/g_\mathrm{N}}$, and the rotation curve is

$$v^2(r) = v_\mathrm{bar}^2(r)\,x\coth x, \qquad x^2 = \frac{a_0\,r}{v_\mathrm{bar}^2(r)}, \tag{11}$$

which is r times the relation (10) at $g_\mathrm{N} = v_\mathrm{bar}^2/r$. Here $v_\mathrm{bar}^2 = v_\star^2 + v_\mathrm{gas}^2 + v_\mathrm{bulge}^2$ is computed from photometry and 21-cm maps exactly as in a Newtonian mass model. The stellar mass-to-light ratio is the only adjustable quantity; a_0 is universal, and there is no halo. For a spherical mass the recipe is exact. For a disk it applies the relation radius by radius to the Newtonian curve of the disk. The two cases meet in the growth of a galaxy. It begins as an aggregate of stars moving at random, with no ordered rotation and no preferred plane. The angular momenta cancel, and the body is a ball. Outside that ball the speed is the point-mass law of the whole baryonic mass, whatever the arrangement inside. Gravity closes on the galactic patch of radius R. The electromagnetic channel reads the same lattice on a larger compactification Λ. Λ is an input, as a_0 is. Inside the patch the gravitational acceleration exceeds its inverse-square value by the fraction $x\coth x - 1$, which lies between 0.31 and 0.32 at $r = R$. Throughout $r \le R$ the electromagnetic fraction stays below $(R/\Lambda)^2/3$, and at $r = R$ the ratio of the gravitational fraction to the electromagnetic one exceeds $\frac{93}{100}(\Lambda/R)^2$. The separation is a centripetal residual of the gravitational channel, radial about the galactic centre. The plane is chosen when the aggregate departs from spherical symmetry and a torque appears. On the residual, the specific angular momentum of a circular orbit increases strictly with radius and tends to zero at the origin, so each value belongs to exactly one radius. Near the centre a small angular momentum still belongs to a small orbit. Past three compactification radii the speed has levelled, and the specific angular momentum grows with the radius.

As the torque carries angular momentum outward, the aggregate expands and keeps a rotation near v_∞. That rotation is the vortex, and the ball spreads into a disk.

At every radius

$$\max\left(v_{\rm bar}^2, \sqrt{a_0\, r\, v_{\rm bar}^2}\right) < v^2 < v_{\rm bar}^2 + \sqrt{a_0\, r\, v_{\rm bar}^2}.$$

The curve therefore vanishes wherever $r\, v_{\rm bar}^2$ does, as at a regular centre. Every baryonic curve with a Keplerian tail, $r\, v_{\rm bar}^2 \to GM_{\rm b}$, flattens at $v_\infty^2 = \sqrt{GM_{\rm b}a_0}$, and outside the baryons the point-mass behaviour above holds with $R = \sqrt{GM_{\rm b}/a_0}$. The curve rises, turns over, and stays flat.

For a Milky-Way exponential disk of $6 \times 10^{10}\, M_\odot$ and scale length 2.6 kpc, Eq. (11) gives about 202 km/s at 4 kpc, 211 at 8, 190 at 15, 179 at 25, and 176 at 60 kpc, while the Newtonian disk alone falls from 188 to 105 km/s between 8 and 25 kpc. A low-surface-brightness disk of $3 \times 10^9\, M_\odot$ and scale length 3 kpc, with $g_{\rm N}$ near $a_0/10$ across the inner disk, rises to about 50 km/s at 2 kpc, 78 at 6 and 87 at 15 kpc, and settles between 83.1 and 83.2 km/s. A dwarf of $4 \times 10^8\, M_\odot$ whose baryons lie inside a ball of radius 0.8 kpc has compactification radius between 0.6816 and 0.6817 kpc, so the quarter-percent flat regime begins between 2.04 and 2.05 kpc. Its asymptotic speed lies between 50.2 and 50.3 km/s. At the surface of the ball the speed lies between 55.2 and 55.3 km/s; by 2 kpc it has fallen to between 50.38 and 50.39 km/s, and at 4 kpc it is back between 50.2 and 50.3 km/s. From that surface to 4 kpc the radius grows by a factor of five, and the specific angular momentum of the circular orbit by a factor between 4.53 and 4.56. On the radial lattice the antipode would lie between 2.141 and 2.142 kpc, inside that flat stretch.

12.7 What the sky can decide

Neighbouring patches may overlap, each faithful near its own centre. If galaxies and clusters are such patches, the excess of the observed acceleration over the baryonic Newtonian value is the dual image lattice, not an unseen mass, and a direct search has nothing particle-like to find. Three features separate this reading from dark halos and from fitted interpolations alike. Outside the baryons the curve approaches $v_\infty = (GM_{\rm b}a_0)^{1/4}$ from above and is flat to a quarter of a percent past $3R$; it neither declines like a Keplerian tail nor rises again. The baryonic Tully–Fisher relation $v_\infty^4 = GM_{\rm b}a_0$ is exact, with no dependence on surface brightness or size. And the relation (10) has a definite transition that lies lower than the empirical one. At $g_{\rm N} = a_0$ the enhancement is 1.31–1.32, where the fit of McGaugh, Lelli and Schombert gives about 1.58. In the galaxies above the dual curve runs up to a tenth below that fit, and the two converge only slowly toward the common v_∞. In star-dominated disks a mass-to-light ratio higher by roughly a fifth absorbs the difference. In gas-dominated dwarfs, whose baryonic curve is fixed by 21-cm data, nothing does, and they decide between the lattice and the fitted relation. The alternation of attraction and repulsion in strong fields, ordered by the interference of equal rapidities, is another regime. It is not the sign of g on this chart.

13 Gravitational Engineering: Coherent Excitation of Dual-Spacetime Rotors

Because $J = \frac{1}{2}\sum_a(\alpha_a^2 - \beta_a^2)$ depends only on the relative rotor, dual-only motion—usual rapidities held fixed—obeys

$$\dot{J} = -\sum_a \beta_a \dot{\beta}_a.$$

Increasing the dual angles cannot raise J. At vanishing dual angle the first variation vanishes, and J drops at second order along a dual ray $\beta = tv$. The admissible cone $\alpha_a, \beta_a \geq 0$, $\alpha_a + \beta_a \leq \pi/2$ is convex, so affine interpolation of dual angles between admissible endpoints of equal usual rapidity remains admissible.

It is worth fixing the target state before asking how to reach it. Call a configuration *gravitationally neutral*, or equivalently *shielded*, when the invariant mismatch vanishes:

$$J = 0, \qquad \text{equivalently} \qquad \sum_a \alpha_a^2 = \sum_a \beta_a^2. \tag{12}$$

The two sectors then hold equal rapidity budgets, and the particle neither attracts nor repels; $J > 0$ is the dual budget falling short, $J < 0$ its overshooting. Neutrality in this sense is a statement about the signed scalar and about nothing else, and it is worth separating from three notions it is easily confused

with. It is not the vacuum: apart from the trivial origin a gravitationally neutral configuration carries strictly positive unsigned mass $M = \sum_a \alpha_a^2$, and the heavier the usual sector, the larger the dual budget neutrality consumes. It is not coincidence of the two rotors: J can vanish while the relative rotor Ω stays away from the identity, so a neutral particle remains internally torsioned. And it is not the absence of structure: where the two rotors share a common scale J vanishes while their interference continues to alternate, so the layered phase of Section 6 survives neutrality untouched. A gravitationally neutral particle is thus not an inert one. It is one whose gravitational and inertial signature, in the sense of Section 5, has been cancelled, while everything else about it remains.

That cone has a dual wall $\beta_a = \pi/2 - \alpha_a$, on which

$$J = \frac{\pi}{2} \sum_a \big(\alpha_a - \pi/4\big).$$

Dual-only J cannot fall below the wall. Shielding $J = 0$ by dual excitation alone is possible if and only if the wall is nonpositive, equivalently $\sum_a \alpha_a \le 3\pi/4$. Repulsion $J < 0$ requires a strictly negative wall.

That budget is global. J is a scalar sum, so dual rapidity parked on a silent axis cancels usual rapidity on a boosted one. The isotropic target $\beta = \alpha$ lies in the cone precisely when $\alpha_a \le \pi/4$ on every axis; dual-only shielding does not require that point. A usual rapidity α on one axis and the same dual rapidity on another is admissible for $0 \le \alpha \le \pi/2$ and has $J = 0$, so even a seed at the cone's edge $\alpha = \pi/2$ can be dual-only shielded. What dual-only cannot shield is a uniformly large boost: three axes at $\pi/3$ exhaust $\sum \alpha \le 3\pi/4$. Uniform seeds make the distinction sharp. At $\alpha = \pi/6$ the equal-scale locus is admissible, massive, and repulsive at the wall. At uniform $\alpha = \pi/3$ no dual-only configuration reaches $J = 0$; a single axis at the same value still does.

The same six numbers define the unsigned mass $M = \frac{1}{2}\sum_a(\alpha_a^2+\beta_a^2)$. Independently of any trajectory,

$$J + M = \sum_a \alpha_a^2, \qquad M - J = \sum_a \beta_a^2.$$

Dual-only motion travels on $J + M = \text{const}$: every drop in J is paid for, one-for-one, by unsigned mass, and $J = 0$ forces the dual budget $\sum \beta^2 = \sum \alpha^2$. When the wall vanishes that shield is unique—the wall itself. When the wall is strictly negative the same mass is shared among a family of dual angles; two silent axes already give two distinct shields. Dual-only J fills the closed interval from the wall up to the pure-usual value. From a pure usual seed one lands at $M = 2M_{\text{seed}}$.

Usual-only motion conserves $M - J$. When the wall is positive, dual excitation cannot cancel J, but unwinding the usual rotor lowers the ceiling $\sum \alpha^2$ itself. Truncating each α_a to at most $\pi/4$ and matching the dual angle always yields a massive equal-scale shield; at uniform $\alpha = \pi/3$ that mixed unwind succeeds where dual-only fails. Dual-only conserves $\sum \alpha^2$, so the vacuum $J = M = 0$ lies on no dual-only orbit except the origin. Mixed control reaches it from every seed by unwinding the usual rotor. Inertial resistance, in the sense of Section 5, tracks J, not M. Synchronising the dual rotor with the usual one can null J while increasing M; reducing both requires walking toward the origin.

The same dual-only ray is not a single return. On a boosted axis the factor γ_{eff} of Section 7 falls as the dual angle rises, and it meets the unboosted value 1 at a unique dual angle β_* whose sine is $\tanh(\alpha/2)$. That meeting precedes the coaxial shield: for every positive usual rapidity at most $\pi/2$ one has $\beta_* < \alpha$, so the admissible equal-scale point, where it exists, already has $\gamma_{\text{eff}} < 1$. Whether the wall itself can still reach 1 is a different comparison. The equation $\cos\alpha = \tanh(\alpha/2)$ has a unique root $\alpha_\dagger$ in $(\pi/3, \pi/2)$; below that root a coaxial dual ray meets $\gamma_{\text{eff}} = 1$ before the wall, and above it every admissible dual angle on the boosted axis remains on the side $\gamma_{\text{eff}} > 1$. Cross-axis compensation, which shields the same seed at the same unsigned mass, leaves the boosted axis unexcited in the dual sector, so that axis keeps the special-relativistic factor $\cosh\alpha$ while the compensating axis contributes the trivial factor 1. Nulling J and restoring the clock are therefore two different points of the same cone.

All of that concerns trajectories. The cone itself, before any trajectory is chosen, already limits what a shield can weigh. The variables that make the limit visible are the axis sum and difference $s_a = \alpha_a + \beta_a$, $d_a = \alpha_a - \beta_a$, in which admissibility reads simply $0 \le s_a \le \pi/2$ and $|d_a| \le s_a$, while the two observables become

$$J = \tfrac{1}{2}\sum_a s_a d_a, \qquad M = \tfrac{1}{4}\sum_a \big(s_a^2 + d_a^2\big).$$

Mass is a squared radius; mismatch is a product of coordinates. On a single axis the cone then forces $(k^2 - s_a^2)(k^2 - d_a^2) \ge 0$ with $k = \pi/2$, that is

$$2k^2\big(\alpha_a^2 + \beta_a^2\big) \le k^4 + \big(\alpha_a^2 - \beta_a^2\big)^2,$$

so an axis can be heavy only by being lopsided. Nothing forbids three axes from being lopsided in opposite senses, and that is precisely the freedom shielding exploits; but the cancellation is itself bounded, because each axis density $\alpha_a^2 - \beta_a^2$ lies in $[-k^2, k^2]$. Three such numbers cannot drive the sum of their pairwise products below $-k^4$, and assembling the two facts already binds the observables in the inequality

$$\pi^2 M \le \frac{5\pi^4}{16} + 4J^2, \qquad \text{equivalently} \qquad M_{\text{norm}} \le \frac{5}{6} + \frac{3}{2}\, J_{\text{norm}}^2. \tag{13}$$

The bound holds on the whole cone. It is sharp throughout the central band $|J| \le \pi^2/8$, which is the only band in which a shield can sit, and it is attained along the wall family

$$\alpha = (\pi/2,\, 0,\, \gamma), \qquad \beta = (0,\, \pi/2,\, \pi/2 - \gamma), \qquad 0 \le \gamma \le \pi/2,$$

for which $J = \frac{\pi}{2}(\gamma - \pi/4)$ and $M = \frac{5\pi^2}{16} + (\gamma - \pi/4)^2$ turn (13) into an identity. Read at neutrality (12), a shielded particle cannot weigh more than

$$M \le \frac{5\pi^2}{16} = \frac{5}{6} \cdot \frac{3\pi^2}{8},$$

five sixths of the cone's own mass ceiling. The remaining sixth of the available mass is closed to neutrality—whatever trajectory is taken, and whatever force drives the dual channel. The heaviest shield in the theory is the midpoint $\gamma = \pi/4$: one axis purely usual, one axis purely dual, the third split evenly, all three on the wall.

The shape of that extremal is as telling as its weight. It is maximally anisotropic: the two sectors do not cancel axis by axis, they exchange roles across axes, and that exchange is what buys the mass. An isotropic shield $\beta = \alpha$ is confined by $\alpha_a \le \pi/4$ to $M \le 3\pi^2/16$, lighter than the optimum by exactly the factor $5/3$. Matching each dual angle to its own usual rapidity—the most natural picture of "aligning the two rotors"—is therefore the inefficient way to shield a heavy particle. Cross-axis compensation is the efficient one.

The parabola (13) is not, however, the envelope of the cone. The estimate on pairwise products that produces it does not use which side of zero the axis densities occupy. Once the signed sum $T = 2J$ is large, that omission is costly: each density already sits near the usual wall $\alpha^2 - \beta^2 = k^2$, and the pairwise products cannot be as negative as $-k^4$. The same observation with the dual wall produces a repulsive companion. The three quadratic pairings together cut the cone down to

$$M \le \frac{5\pi^2}{16} + \frac{1}{\pi^2} \min\bigl(4J^2,\, (2J - \tfrac{\pi^2}{2})^2,\, (2J + \tfrac{\pi^2}{2})^2\bigr), \tag{14}$$

or equivalently

$$M_{\text{norm}} \le \frac{5}{6} + \frac{1}{6} \min\bigl(9J_{\text{norm}}^2,\, (3J_{\text{norm}} - 2)^2,\, (3J_{\text{norm}} + 2)^2\bigr).$$

Every pair with $|J| \le M$ beneath this envelope is realised by an admissible configuration, and no pair above it is. The kinematic domain of gravitational control is therefore this region and nothing else.

Its geometry is three-lobed. The central lobe is the shield parabola, running from $J = -\pi^2/8$ to $J = \pi^2/8$ and meeting the absolute mass ceiling $M = 3\pi^2/8$ at both ends. Beyond those ends two axes lock to the same sector at the cone's edge, and the remaining wall traces a parabola centred not at $J = 0$ but at $|J| = \pi^2/4$. The envelope therefore falls back to the shield value $M = 5\pi^2/16$ at $|J| = \pi^2/4$, then climbs the outer lobe to the three-axis lock at $|J| = 3\pi^2/8$, where the mass ceiling is met again. The outer arcs are attained by two pure-usual axes plus a wall, and by the usual–dual swap of that family. Neutrality remains the expensive crossing of the domain; there is a second, cheaper, local minimum where two axes are already locked and the third is itself a shield.

These families are the whole of the envelope. An admissible configuration meets it precisely when every axis lies on the wall and at least two axes are purely usual or purely dual. In the open central band the locked pair occupies opposite sectors, and J fixes how the remaining axis is split: each assignment of the three roles determines one configuration, and the heaviest shield is the one that halves the free axis. On an open outer arc the locked pair belongs to one sector, and J again fixes the rest. The wall reads the dual angles off the usual rapidities, so the boundary has no internal freedom. Dual-only motion holds those rapidities fixed and therefore meets the envelope at most once; sliding along the ceiling, from a light shield toward a two-axis lock or across neutrality, has to move the usual rotor.

At the top of the mass range the freedom vanishes altogether. A configuration attaining $M = 3\pi^2/8$ must have every axis either purely usual, $(\alpha_a, \beta_a) = (\pi/2, 0)$, or purely dual, $(0, \pi/2)$; at maximal radius

the cone admits no intermediate. Such a configuration's mismatch is a sum of three quanta $\pm\pi^2/8$, so $|J_{\mathrm{norm}}| \geq 1/3$. The heaviest configurations the theory admits are not merely hard to shield: they are not gravitationally neutral at all, and nothing can make them so without first making them lighter. They occur at the four points where the envelope meets the mass ceiling—the two-axis locks at $|J| = \pi^2/8$ and the three-axis locks at $|J| = 3\pi^2/8$—and nowhere else.

The asymmetry between the two directions deserves saying plainly, since it inverts the intuition that repulsion should be the expensive state. The ceiling in (13) is a ceiling on neutrality, not on gravity. Full repulsion costs nothing: the pure-elliptic configuration $\alpha = 0$, $\beta_a = \pi/2$ carries the maximal mass $3\pi^2/8$ together with the extreme value $J_{\mathrm{norm}} = -1$, and its attractive mirror does the same. Attraction and repulsion are available at the mass ceiling; what is obstructed is the crossing between them, and, once two axes have locked, a second crossing at $|J| = \pi^2/4$. The design constraint of a control programme is therefore not how much dual excitation can be supplied, but which lobe of the domain one is willing to occupy, and how heavy an object one is willing to leave gravitationally visible.

Brought back to the seed, the two budgets prove complementary rather than redundant. A usual seed that dual excitation alone can shield obeys the linear condition $\sum_a \alpha_a \leq 3\pi/4$, and the shield it reaches carries $M = \sum_a \alpha_a^2$; the ceiling therefore imposes the quadratic condition $\sum_a \alpha_a^2 \leq 5\pi^2/16$ on that same seed.

An electromagnetic coupling, a resonance frequency, or a laboratory protocol is not forced by this geometry. What the geometry does force, once a laboratory reading of a dual-only increment is chosen, is the kinematics of J. Superposing Faraday coefficients onto the mismatch, and substituting $\theta_a \mapsto \theta_a + eA_a$ in the dual rotor, occupy the same six-space: they coincide when the increment is magnetic and $B = eA$, and they part company under a constant shift of A. The remainder of this section records those identities, and then a proposed helicity-odd drive that is not a consequence of either reading.

13.1 Theoretical Coupling of Circularly Polarized Electromagnetic Fields to Dual Rotors

The electromagnetic field occupies the six-generator space of Section 11, partitioned as in (6). Laboratory time reversal $(E, B) \mapsto (E, -B)$ remains a distinct involution from both duality and the usual–dual parameter swap.

A Faraday increment added to the particle's mismatch bivector is superposition in that six-space. A purely magnetic increment holds the usual rapidities fixed, so it is dual-only, conserves $J + M$, and cannot raise J when the existing dual angles and the increment are nonnegative. A purely electric increment is usual-only. The circular wave of that section is mixed. It is null at every phase, and every linear Faraday coefficient has vanishing four-phase mean, so

$$\langle F_{\mathrm{usual}} \rangle = \langle F_{\mathrm{dual}} \rangle = 0.$$

There is in particular no mean dual field proportional to $\sigma E_0^2 K$. Superposition of the wave onto a background torsion does not shift the mean of J. What is helicity-odd and constant is the bilinear Poynting component $(E \times B)_z = \sigma E_0^2$ of that section. Linear polarisation carries a nonnegative Poynting component and no helicity sign. The Faraday mix cannot create J from the wave. A first-order rest sandwich of the dual summand vanishes for every Faraday field, circular waves included. The rest sandwich of the wave itself is the oscillating electric kick, whose four-phase mean vanishes; a velocity along the beam produces a transverse kick with no longitudinal component (7). Axis projectors $P_{L,R}$ likewise fail to extract helicity.

The dual-rotor substitution $\theta_a \mapsto \theta_a + eA_a$ is the same dual-only increment, read as a potential: it equals magnetic superposition of eA. Instantaneously

$$J = J_0 - e\,\beta \cdot A - \tfrac{1}{2} e^2 A^2. \tag{15}$$

For a Coulomb-gauge circular potential of helicity $\sigma^2 = 1$, $A^2 = (E_0/\omega)^2$ is constant and independent of σ; the four-phase mean of J therefore drops by the helicity-even ponderomotive $\frac{1}{2} e^2 (E_0/\omega)^2$. A linearly polarised potential of the same E_0 drops the mean by half that amount. Cycle-averaged $\dot{J}$ vanishes: the shift is a displacement of the mean, not a secular drift. Two potentials that differ by a constant—both would have vanishing Faraday field—yield different values of J; the substitution is not Maxwell-gauge invariant, and that is where the two readings part company. From a vanishing dual seed a zero-mean circular A makes a dual angle negative, and so leaves the admissible cone. An admissible realisation therefore requires a dual bias no smaller than the oscillation, together with remaining room

to the wall. Magnetostatic dual excitation is the chamber-preserving case of the same kinematics, and the $1/\omega^2$ suppression of the ponderomotive drop is why that channel prefers slow magnetic excitation to a high-frequency wave.

Neither reading is helicity-odd in J. A helicity-dependent drive of the particle mismatch is therefore not a local scalar of F, nor a linear superposition, nor a first-order rest sandwich of F_{dual}, nor an axis-chirality projection, nor a beam-direction Lorentz kick, nor the dual-rotor substitution $\theta \mapsto \theta + eA$. Any such drive requires extra structure—a preferred axis, a potential, or the particle's dual phase. Within the classical quadratic span the helicity-odd candidate remains $(E \times B)_z = \sigma E_0^2$. In the model below that bilinear is the source of a dual-phase drive, with right-handed polarisation ($\sigma = +1$) reinforcing attractive mismatch ($J > 0$) and left-handed polarisation ($\sigma = -1$) opposing it. That identification remains a model.

If a dual-channel generalised acceleration u is added to $\ddot{\theta}$ in the written particle action of Section 16, the mismatch is a double integrator

$$\ddot{\delta} = -u,$$

with no restoring frequency. Dual-only kinematics is not a free solution of that action: holding $\ddot{\phi} = 0$ uniquely requires the usual-channel force $v = m(\phi - \theta)$. Realising the dual-only trajectory $\theta = \theta_0 + eA$ further requires $u = e\ddot{A} + m(\phi - \theta)$: the coupling $e\ddot{A}$ is not by itself the Euler–Lagrange dual source unless the mismatch already vanishes. The neighbouring same-sign oscillator instead obeys $\ddot{\delta} + 2m\delta = -u$. Resonance remains extra structure, not an identity of the written action.

The coupling in the working model is written as a path-ordered exponential

$$R_{\mathrm{dual}} \to \mathcal{P} \exp\left(\frac{1}{2} \int (\theta_a + eA_a)\, \Gamma_a \, ds \right),$$

where the coupling e absorbs the generator scale, and the dual generators Γ_a mediate the interaction. The phase-lock

$$\dot{\theta} = \kappa \sigma E_0^2 \sin(\omega t - \theta + \delta_\sigma)$$

is a model of that extra structure, not a consequence of the admissible-cone kinematics or of the written Euler–Lagrange system.

13.2 Resonance Frequencies and the Role of Superconductors

The intrinsic dual-spacetime scale is Planckian ($\sim 10^{43}$ Hz). Macroscopic coherence ($N \sim 10^{26}$ particles) is proposed to redshift an effective frequency to 10^{12}–10^{15} Hz. Superconductors would further reduce it via Josephson plasma resonances and collective phase locking, achieving tunable modes in the GHz–THz range accessible to current technology. That reduction is part of the working model; the written action itself supplies no Compton oscillator.

13.3 Proposed Experimental Protocol and Predictions

A microgram-scale test mass in a high-Q superconducting resonator is illuminated with tunable circularly polarized radiation (0.1–10 THz). Expected signatures, if the dual-channel coupling exists, include helicity-dependent changes in apparent weight or inertial mass ($\Delta m/m \gtrsim 10^{-6}$), with left-handed polarization producing reduction and right-handed enhancement. A resonant peak would test the extra phase-lock structure, not the dual-only geometry of J, and not the helicity-even ponderomotive of (15). The algebraic channel already prefers slow magnetic dual excitation: a high-frequency circular wave is the wrong regime for that drop, and from a vanishing dual seed it is not even an admissible trajectory. Success of the chiral protocol would therefore be evidence for the extra model, not a confirmation of the dual-only increment.

If the model is right, gravity is not merely controllable but chirally controllable, with helicity serving as the switch between attraction and repulsion. The admissible cone's image in the (J, M)-plane is known exactly: a three-lobed domain bounded by (14). Dual-only kinematics already say which trajectories stay inside it. The dual-only increment already supplies a helicity-even, low-frequency channel along those trajectories. Whether laboratory electromagnetism supplies a helicity-odd dual force is a conjecture, to be decided in the laboratory.

14 Baryon Asymmetry and the Sign of the Vacuum Excess

The cosmic baryon asymmetry is a small number,

$$\eta \equiv \frac{n_B - n_{\bar{B}}}{n_\gamma} \approx 6 \times 10^{-10}.$$

The dual rotors determine a different quantity. A chiral excess of the torsionless vacuum places one unsigned mass on two opposite values of the Killing scalar, and the ratio of the scalar to that mass is maximal. The yield η belongs to a later stage of the history.

14.1 Equal mass, opposite sign

On one axis the scalars are $J = \frac{1}{2}(\alpha^2 - \beta^2)$ and $M = \frac{1}{2}(\alpha^2 + \beta^2)$. The torsionless vacuum is the common zero of the two rapidities. Equal rapidities $\alpha = \beta = \varphi$ form another state: J vanishes and the unsigned mass is φ^2. Whenever $\varphi \neq 0$ the two rotors disagree, and the relative rotor differs from the identity. A balanced equilibrium of nonzero rapidity is massive. Read as torsionless, the empty early universe sits at the common zero.

From that zero, let a rapidity δ run along a direction p in the three axes, normalised by $\sum_a p_a^2 = 1$. If the excess occupies the boost generators,

$$J = M = \frac{\delta^2}{2}. \tag{16}$$

If it occupies the rotation generators,

$$J = -\frac{\delta^2}{2}, \qquad M = \frac{\delta^2}{2}. \tag{17}$$

Along a general direction the same relations hold with δ^2 replaced by $\delta^2 \sum_a p_a^2$, so $J = M$ on the boost side and $J = -M$ on the rotation side. Swapping the sectors exchanges the two lines and leaves M fixed. The dual map, acting on the torsion bivector, performs the same exchange up to the sign of the rapidity and reverses J. A second application multiplies every element by -1 and returns each sector to itself. Replacing δ by $-\delta$ changes neither scalar, since both are quadratic. Whenever $\delta \neq 0$ and $p \neq 0$,

$$\frac{J}{M} = \pm 1.$$

A pure chiral seed is gravitationally maximal. The observed η lies many orders of magnitude below this ratio.

The same shift, placed on a one-axis balance of rapidity φ, produces

$$J = \varphi\delta + \frac{\delta^2}{2}, \qquad M = \varphi^2 + \varphi\delta + \frac{\delta^2}{2}. \tag{18}$$

The two signs of δ then differ in mass by $2\varphi\delta$. The cross term in J, and this mass difference, drop out for every δ precisely when $\varphi = 0$. The quadratic law (16)–(17) is the response of the vacuum. A massive equilibrium answers with a term linear in the excess.

14.2 The bridge and the yield

Spatial inversion fixes the time axis and the null direction, and it reverses the three spatial axes. The pseudoscalar changes sign. Each boost generator changes sign. Each rotation generator stays fixed. The bridge $X \mapsto Xi$ is therefore parity-odd, while the sectors themselves are preserved: the image of a boost excess δ is the boost excess $-\delta$, which carries the same J, and a rotation excess stays where it stands. The sign of J records which factor of the Lorentz span the excess occupies.

On $\mathbb{C}^2$ the dual rotor is unitary, and its determinant equals 1 at every rapidity. The argument of that determinant vanishes. The asymmetry has its place in the sign of J, at one common unsigned mass.

On the radial chart the positive sign is attractive and the negative sign is repulsive, and the repulsive depth is bounded. Both signs are available at the mass $M = \delta^2/2$. A configuration that binds occupies the positive sign. Its partner is the sector swap, carrying $J = -M$ and the same M.

What Sakharov required of a cosmological history is a violation of baryon number, a violation of C and CP, and a departure from thermal equilibrium, enough to freeze a number density. The geometry

that precedes that history is the parity-odd bridge between two sectors of equal mass and opposite gravitational character. A later stage that froze a density proportional to δ^2 would meet the observed η for a δ of order 10^{-5}. That proportionality is a hypothesis about the freeze-out, laid over a scalar that the vacuum excess has already fixed.

15 Discrete Time and Compactified Torsion Scalar

The rejection of the spacetime continuum hypothesis demands a discrete structure at the fundamental scale. The intrinsic duality encoded in the biquaternion algebra—particularly the pseudoscalar i and the dual map $X \mapsto Xi$ that reverses the temporal arrow while preserving the Minkowski norm—compactifies the dual spacetime intrinsically, while leaving the usual spacetime non-compact.

We therefore discretize only the dual-spacetime rotation angle:

$$\theta \in \frac{2\pi}{N}\mathbb{Z},$$

where $N \gg 1$ is a very large integer determining the compactification scale of the dual spacetime (typically proportional to the particle's rest mass in Planck units). On that reading the nonzero normalized floor is sixteen thirds of $Gm^2/\hbar c$, the gravitational scale compared with the electromagnetic constant in Section 9. The usual-spacetime rapidity remains continuous:

$$\phi \in \mathbb{R}.$$

This asymmetry is required for theoretical consistency:

- The dual spacetime is elliptic ($\hat{q}^2 = -1$), inherently compact, and its discretization directly reflects the 4π-periodic topology of S^3 parameterized by R_{dual}. This provides the finite phase volume essential for resolving the Newton–Wigner localization problem and yielding natural quantum discreteness.
- The usual spacetime is hyperbolic ($\hat{p}^2 = +1$), intrinsically non-compact, so continuity of ϕ preserves the full Lorentz group in the macroscopic limit.

The torsional mismatch rotor $\Omega = R_{\text{usual}}^{\dagger} R_{\text{dual}}$ thus combines a continuous usual component with a discrete dual component. The scalar J is bounded by $|J| \leq \frac{3\pi^2}{8}$ only on admissible configurations. Discretizing the dual angle alone does not imply the bound: a principal-branch condition without non-negative rapidities admits counterexamples, and the continuous usual-sector rapidity is unbounded. Continuum-induced singularities are therefore excluded only after the admissible cone is imposed. On the discrete torus $(\mathbb{Z}/N\mathbb{Z})^6$ the rotor image is finite. Nonzero winding of a scaled lattice configuration is incompatible with real-scale admissibility. When $4 \mid N$ the lattice attains the extremals $J_{\text{norm}} = \pm 1$; otherwise the admissible bound is strict.

A critical issue arises in the effective Lorentz factor across torsional layers, which typically appears as a synchronization function of the form

$$\gamma_{\text{eff}} \propto D_N(\delta\theta) \equiv \frac{\sin(N\delta\theta/2)}{N\sin(\delta\theta/2)},$$

where $\delta\theta$ is the torsional mismatch angle and D_N is the (normalized) Dirichlet kernel. The denominator vanishes precisely on the lattice $\delta\theta \in 2\pi\mathbb{Z}$.

At every such point the numerator $\sin(N \cdot k\pi)$ also vanishes (for any integer N, prime or composite), so the apparent singularity is in fact a removable $0/0$ of Dirichlet-kernel type. The standard L'Hôpital evaluation gives the finite limit

$$\lim_{\delta\theta \to 2\pi k} D_N(\delta\theta) = \pm 1 \qquad (k \in \mathbb{Z}),$$

so the synchronization function is bounded on the entire compact lattice and the classical continuum limit $N \to \infty$ is recovered without pathology.

Thus the Dirichlet kernel is bounded for any large positive integer N; primality is not required. Filling the removable singularities by the L'Hôpital value yields $|D_N(\theta)| \leq 1$, with $D_N = \pm 1$ at every lattice point. The relation between N and particle mass spectra, and the zeta-regularized product $\prod_p p \to 4\pi^2$, remain open. The geometric reading of $4\pi^2$ as the square of the dual-rotor period 2π is a consistency remark, not a derivation of mass spectra.

16 Dual-Rotor Dynamics and the Geometric Origin of the de Broglie Phase

The mismatch scalar J is proposed as a gravitational action density in the classical field theory. The particle action uses a different quantity. The de Broglie identification below remains a proposal.

The potential of that action is the stiffness of the lag, $V = \frac{m}{2}\sum_a(\phi_a - \theta_a)^2$. Taking the rest mass m as its coefficient is a modelling choice, not a derivation of m from the algebra. On one axis, with σ the sum of the rapidities and δ their difference,

$$V = 2m\Big(M - \frac{\sigma^2}{4}\Big), \tag{19}$$

while $J = \frac{1}{2}\delta\sigma$. For nonzero m the two agree up to that factor precisely when the lag vanishes or the dual angle vanishes, and both are nonzero together only on the pure-usual ray. There the dual angles vanish and $J = \frac{1}{2}\sum_a \phi_a^2$. That ray is not the definition of J.

We consider the written effective action for the intrinsic rotor angles of a single particle:

$$S_{\text{particle}} = \int \left[\frac{1}{2}\sum_{a=1}^{3}\dot{\phi}_a^2 - \frac{1}{2}\sum_{a=1}^{3}\dot{\theta}_a^2 - \frac{m}{2}\sum_{a=1}^{3}(\phi_a - \theta_a)^2\right] dt.$$

The opposite signs of the kinetic terms are part of the written model; they are not deduced from unitarity of the rotors.

The Euler–Lagrange equations of this Lagrangian are

$$\ddot{\phi} = -m(\phi - \theta), \qquad \ddot{\theta} = -m(\phi - \theta).$$

The mismatch $\delta = \phi - \theta$ is therefore free: $\ddot{\delta} = 0$. On one axis the integral is the affine lag $\delta(t) = \delta_0 + \nu t$ together with the cubic common rapidity

$$\sigma(t) = \sigma_0 + \dot{\sigma}_0\, t - m\delta_0\, t^2 - \frac{m\nu}{3}\, t^3,$$

and the Jacobi integral along that motion equals the initial value $\frac{1}{2}\nu\dot{\sigma}_0 + \frac{m}{2}\delta_0^2$. The written action does not yield the oscillator $\ddot{\delta} + m\delta = 0$. A neighbouring system sometimes written alongside it,

$$\ddot{\phi} = m(\phi - \theta), \qquad -\ddot{\theta} = m(\phi - \theta),$$

makes the mismatch runaway, $\ddot{\delta} = 2m\delta$. For $m > 0$ that runaway is the hyperbolic motion $A\cosh(\sqrt{2m}\,t) + B\sinh(\sqrt{2m}\,t)$, unbounded for a nonzero seed, and is likewise not an oscillator.

An oscillator does appear if both kinetic signs are taken positive. The modified density $L = \frac{1}{2}\dot{\phi}^2 + \frac{1}{2}\dot{\theta}^2 - \frac{m}{2}(\phi - \theta)^2$ has Euler–Lagrange equations $\ddot{\phi} = -m(\phi - \theta)$ and $\ddot{\theta} = m(\phi - \theta)$, hence

$$\ddot{\delta} + 2m\delta = 0.$$

This is a separate working model, not the written action. Its frequency is $\sqrt{2m}$ (in natural units $\hbar = c = 1$). That line and the Compton frequency m meet only at the isolated values $m = 0$ and $m = 2$. For a free particle in uniform motion the usual rapidity is modelled as evolving at constant rate $\dot{\phi}_a = \gamma v_a/c$. The general solution is

$$\delta(t) = A\cos(\sqrt{2m}\,t) + B\sin(\sqrt{2m}\,t),$$

with $\delta(t)^2 \le A^2 + B^2$ and with Jacobi integral $\frac{1}{4}\dot{\sigma}_0^2 + \frac{m}{2}(A^2 + B^2)$. In the low-energy, non-relativistic regime, rapid Compton-frequency oscillations are assumed to average to zero, leaving a slow linear accumulation of phase proportional to the proper time. Restoring $\hbar$ and c, the accumulated phase is conjectured to be

$$\int \delta\phi_a\, dt \propto \frac{Et - \mathbf{p}\cdot\mathbf{x}}{\hbar},$$

the de Broglie phase of the matter wave. Collective averaging of this particle-intrinsic action over many particles is proposed as a macroscopic reading of the mismatch. That averaging identifies neither the stiffness potential with J nor J with the Weitzenböck density T. The written action does not furnish a harmonic bridge to quantum theory; the same-sign model is the only oscillator among the three systems compared here. An arbitrary potential of the mismatch does not change the opposite-sign conclusion:

the lag stays free, and a constant lag is inertial only where the slope vanishes, so the harmonic stiffness accelerates any nonzero lag. A quartic well off alignment freezes a nonzero lag only when the kinetic signs agree, and the flat rate is then initial data, not a mass for W or Z.

A dual-channel generalised acceleration u added to $\ddot{\theta}$ does not restore a frequency in the written action: $\ddot{\delta} = -u$ and $\ddot{\sigma} = -2m\delta + u$. Those identities assume both channels free. Holding $\ddot{\phi} = 0$ is a constraint. If a usual-channel force v is admitted together with u, the written Jacobi integral changes at the rate $v\dot{\phi} - u\dot{\theta}$, and $\ddot{\phi} = 0$ uniquely fixes $v = m(\phi - \theta)$. Dual-only kinematics is therefore not a free solution: leftover mismatch with nonzero rest mass drags the usual sector, and realising the dual-only slice requires that unique constraint force. When the usual rapidity is instantaneously at rest, $\dot{\phi} = 0$, the constraint contributes no power; only the dual pairing $-u\dot{\theta}$ remains. The neighbouring oscillator instead obeys $\ddot{\delta} + 2m\delta = -u$. Whether laboratory fields supply such a u is the modelling question of Section 13, not a consequence of these identities.

17 Towards Quantum Gravity and Full Unification

The classical formulation presented thus far treats gravity as torsional mismatch between the particle-intrinsic usual and dual spacetimes, encoded in $\mathbb{H} \otimes_{\mathbb{R}} \mathbb{H} \cong \mathrm{Cl}(3,1)$. The proposed action uses the parameter-space scalar J; it is not identified with the Einstein–Hilbert integral.

The dual rotor $R_{\mathrm{dual}} = \exp(\theta/2\,\hat{q})$ is the spin-1/2 rotation operator of SU(2). Dual-sector phases θ therefore parameterize quantum states. Represented on $\mathbb{C}^2$ by $\exp\big(\sum_a (\theta_a/2)(-i\sigma_a)\big)$, the dual rotor is unitary with determinant 1, hence an element of SU(2). Due to the compactness of the dual spacetime ($\theta \bmod 4\pi$), these phases live on a compact manifold: each particle's state $|\psi\rangle$ corresponds to the unitary phase part of R_{dual}. A Hilbert-space completion of a $\mathrm{Cl}(3,1)$ minimal left ideal is not constructed here.

Position and momentum operators, and uncertainty relations among them, are not derived. Singularities are excluded on the admissible cone by the bound of Section 4; the quasi-horizon of Section 6.2 is a reading of that bound on the saturated Schwarzschild chart. An explicit Hilbert space of $\mathrm{Cl}(3,1)$ spinors, and a complete quantization of the mismatch dynamics, remain open.

Quantum gravity is therefore not a separate theory: it is the dual-rotor kinematics of the same 16-dimensional algebra, still incomplete as a Hilbert-space construction.

18 Conclusions

The century that began with Einstein taught us to speak of gravity as the curvature of a continuum. That speech is exact as a macroscopic bookkeeping, and it remains so. What it does not say is why a continuum should curve, nor why the same phenomenon should appear as inertia under acceleration. The present work has refused the continuum as a primitive. There is no autonomous stage. Each massive particle carries, in the sixteen-real-dimensional biquaternion algebra isomorphic to $\mathrm{Cl}(3,1)$, a pair of Minkowski sectors—the usual and the dual. A map between them reverses the arrow of time and leaves the Minkowski norm untouched. Gravity is the torsion of that pair: the mismatch of two rotors that belong to the particle itself.

The relative rotor is the identity precisely when the two rotors coincide. A Lorentz-invariant scalar J measures the expressed mismatch, the admissible point closest to the sum of the source bivectors. That sum itself has no ceiling. J is bounded on the admissible cone, and vanishing J does not force the rotors to agree. On the diagonal radial chart the teleparallel scalar is $T = -2(A + rA')/r^2$. The vacuum equation $\frac{d}{dr}(rA) = 1$, which is $G^t{}_t = G^r{}_r = 0$, selects the exterior Schwarzschild factor and freezes T at $-2/r^2$, the value of the flat spherical frame. The mismatch $J = \frac{1}{2}\varphi^2$ on that chart is a different scalar: it tracks the Newtonian potential and is not determined by T. A boost of the inertial frame in one fixed plane, of any rapidity profile, leaves both the Minkowski metric and the teleparallel scalar untouched, even where the torsion does not vanish, and J remains free. A real radial boost has positive J; repulsive layers belong to the elliptic sector and are of bounded depth. Where the two rotors share a common scale, J vanishes while their interference still alternates, so the layers of strong-field gravity are ordered by that interference and not by J. The same algebra houses electromagnetism: Faraday coefficients occupy the six-generator space of the dual rotor, and the sandwich increment is a Lorentz 4-force. A spatial chirality axis splits that six-space into an $\mathfrak{so}(2,1)$ stabilizer and a complement that moves the axis. A perpendicular dual rotation exchanges the two projectors at angle π, and the stabilizer's only compact direction admits no nontrivial mix with the parallel electric generator. Compact SU(2) is the dual rotor, not a chiral gauge symmetry. On a three-sphere chart the circular speed is $v^2 = (GM/R)f(r/R)$. The

shape f has a unique minimum on $(0, \pi/2)$, and from one compactification radius to six-fifths of it, v^2 stays within a tenth of that minimum. The radial force remains center-directed through the equator, where the negative well ends, before $r = 2R$. None of this requires a background manifold.

We have proposed that the layered phase is the strong force; that galaxies are patches of the compact dual sector, whose image lattice therefore accounts for flat rotation curves; that coherent excitation of the dual sector might engineer gravity; that a vacuum chiral excess assigns one unsigned mass to both signs of J, with binding on the positive sign, while the yield η lies outside that assignment; that Maxwell's equations remain unrecovered from the same six-space, and that a harmonic stiffness does not select a rotor rigidity away from alignment; that a mismatch oscillator is the de Broglie phase; and that a Hilbert-space quantization of the dual rotor completes the theory. These are readings of the geometry. Dual-only motion on the admissible cone cannot raise J. It conserves $J + M$, so every dual-only shield from a pure usual seed doubles unsigned mass, and it reaches $J = 0$ precisely when the dual wall is nonpositive. On a boosted axis a coaxial dual-only ray restores the clock factor γ_{eff} to 1 strictly before it cancels J; a cross-axis shield of equal mass leaves that clock at $\cosh \alpha$. Independently of any trajectory the cone's image in the (J, M)-plane is the three-lobed envelope (14): the shield parabola is sharp only for $|J| \leq \pi^2/8$, neutrality is capped at five sixths of the maximal mass and is reached only by a maximally anisotropic wall configuration, and at maximal mass every axis is purely usual or purely dual—while full repulsion pays no such penalty. The envelope is a lock—every axis on the wall, at least two of them purely usual or purely dual—so dual-only motion meets it at most once and cannot travel along it. Holding the usual rotor fixed against the written potential requires a constraint. A dual-only increment of the dual angles—magnetic Faraday superposition, or the substitution $\theta \mapsto \theta + eA$—cannot raise J against a nonnegative seed; along a circular wave the Faraday reading leaves mean J unmoved, while the potential reading drops it by a helicity-even ponderomotive that is not Maxwell-gauge invariant and that leaves the cone from a vanishing dual seed. Whether laboratory electromagnetism supplies a helicity-odd dual force—and whether gravity is therefore chirally controllable—remains a conjecture.

A balanced massive seed is not emptied by either rapidity budget alone, and on the aligned flow its mass stands still only when the common rate vanishes. A hydrogen ionisation does not pay the electron rest energy, and the first Coulombic node is not a wall.

What can already be asked of the sky is modest and sharp. In place of a true horizon one expects a finite redshift floor. At the centre of a spherically symmetric torsion star, gravitational time dilation should vanish. If a galaxy is a patch of the compact dual sector, its rotation curve is the baryonic Newtonian curve times $x \coth x$, with $x^2 = a_0 r / v_{\text{bar}}^2$, rather than the trace of unseen mass. The vortex is the rotation supported by the centripetal residual of that patch, and inside the patch the electromagnetic departure from inverse-square stays below $(R/\Lambda)^2/3$. A solar mass at one astronomical unit exceeds its Newtonian rapidity by a fraction below $1/(4 \times 10^7)$ and, through the dual images, the inverse square by a fraction below 10^{-8}; at the Milky-Way compactification radius the rapidity fraction stays below 10^{-6} beside a dual excess between 0.31 and 0.32, and tying the patch radius to the enclosed mass makes the acceleration relation strictly subhomogeneous and strictly subadditive. The curve flattens from above at $v_\infty^4 = G M_{\text{b}} a_0$, between 175 and 176 km/s for the Milky-Way stand-in mass. A spherical dwarf of $4 \times 10^8\, M_\odot$ is already in that asymptote, between 50.2 and 50.3 km/s, once the orbit lies outside its ball. Carried from the surface of a 0.8 kpc ball out to 4 kpc, the specific angular momentum of the circular orbit rises by a factor between 4.53 and 4.56, and the speed returns to the same asymptote. At $g_{\text{N}} = a_0$ the enhancement is $\coth 1 \approx 1.31$, below the empirical 1.58. The ratio of successive barrier radii is a pure number, the same at nuclear and at stellar scales, because the characteristic length cancels from it.

If this ontology is sound, then the geometry of the world is not prior to matter. It is an attribute of particles, as spin and charge are attributes, and the apparent curvature of the heavens is the torsion of each particle's own dual rotors. The continuum remains a useful fiction of the large. What remains is not to curve an empty stage more ingeniously, but to learn, from the algebra already in hand, how the paired rotors of matter compose the world we see.

A Multiplication Table

Table 1: Biquaternion Multiplication Table

	j	kI	kJ	kK	iI	iJ	iK	I	J	K	k	jI	jJ	jK	i
j	-1	$+iI$	$+iJ$	$+iK$	$-kI$	$-kJ$	$-kK$	$+jI$	$+jJ$	$+jK$	$+i$	$-I$	$-J$	$-K$	$-k$
kI	$-iI$	$+1$	$-K$	$+J$	$-j$	$+jK$	$-jJ$	$-k$	$+kK$	$-kJ$	$-I$	$+i$	$-iK$	$+iJ$	$+jI$
kJ	$-iJ$	$+K$	$+1$	$-I$	$-jK$	$-j$	$+jI$	$-kK$	$-k$	$+kI$	$-J$	$+iK$	$+i$	$-iI$	$+jJ$
kK	$-iK$	$-J$	$+I$	$+1$	$+jJ$	$-jI$	$-j$	$+kJ$	$-kI$	$-k$	$-K$	$-iJ$	$+iI$	$+i$	$+jK$
iI	$+kI$	$+j$	$-jK$	$+jJ$	$+1$	$-K$	$+J$	$-i$	$+iK$	$-iJ$	$-jI$	$-k$	$+kK$	$-kJ$	$-I$
iJ	$+kJ$	$+jK$	$+j$	$-jI$	$+K$	$+1$	$-I$	$-iK$	$-i$	$+iI$	$-jJ$	$-kK$	$-k$	$+kI$	$-J$
iK	$+kK$	$-jJ$	$+jI$	$+j$	$-J$	$+I$	$+1$	$+iJ$	$-iI$	$-i$	$-jK$	$+kJ$	$-kI$	$-k$	$-K$
I	$+jI$	$-k$	$+kK$	$-kJ$	$-i$	$+iK$	$-iJ$	-1	$+K$	$-J$	$+kI$	$-j$	$+jK$	$-jJ$	$+iI$
J	$+jJ$	$-kK$	$-k$	$+kI$	$-iK$	$-i$	$+iI$	$-K$	-1	$+I$	$+kJ$	$-jK$	$-j$	$+jI$	$+iJ$
K	$+jK$	$+kJ$	$-kI$	$-k$	$+iJ$	$-iI$	$-i$	$+J$	$-I$	-1	$+kK$	$+jJ$	$-jI$	$-j$	$+iK$
k	$-i$	$-I$	$-J$	$-K$	$+jI$	$+jJ$	$+jK$	$+kI$	$+kJ$	$+kK$	-1	$-iI$	$-iJ$	$-iK$	$+j$
jI	$-I$	$-i$	$+iK$	$-iJ$	$+k$	$-kK$	$+kJ$	$-j$	$+jK$	$-jJ$	$+iI$	$+1$	$-K$	$+J$	$-kI$
jJ	$-J$	$-iK$	$-i$	$+iI$	$+kK$	$+k$	$-kI$	$-jK$	$-j$	$+jI$	$+iJ$	$+K$	$+1$	$-I$	$-kJ$
jK	$-K$	$+iJ$	$-iI$	$-i$	$-kJ$	$+kI$	$+k$	$+jJ$	$-jI$	$-j$	$+iK$	$-J$	$+I$	$+1$	$-kK$
i	$+k$	$-jI$	$-jJ$	$-jK$	$-I$	$-J$	$-K$	$+iI$	$+iJ$	$+iK$	$-j$	$+kI$	$+kJ$	$+kK$	-1

Table 2: Cl(3,1) Variant Multiplication Table ($e_0, e_1, e_2, e_3 \mapsto 0, 1, 2, 3$)

	0	1	2	3	01	02	03	32	13	21	123	032	013	021	0123
0	$-$	$+01$	$+02$	$+03$	-1	-2	-3	$+032$	$+013$	$+021$	$+0123$	-32	-13	-21	-123
1	-01	$+$	-21	$+13$	-0	$+021$	-013	-123	$+3$	-2	-32	$+0123$	-03	$+02$	$+032$
2	-02	$+21$	$+$	-32	-021	-0	$+032$	-3	-123	$+1$	-13	$+03$	$+0123$	-01	$+013$
3	-03	-13	$+32$	$+$	$+013$	-032	-0	$+2$	-1	-123	-21	-02	$+01$	$+0123$	$+021$
01	$+1$	$+0$	-021	$+013$	$+$	-21	$+13$	-0123	$+03$	-02	-032	-123	$+3$	-2	-32
02	$+2$	$+021$	$+0$	-032	$+21$	$+$	-32	-03	-0123	$+01$	-013	-3	-123	$+1$	-13
03	$+3$	-013	$+032$	$+0$	-13	$+32$	$+$	$+02$	-01	-0123	-021	$+2$	-1	-123	-21
32	$+032$	-123	$+3$	-2	-0123	$+03$	-02	$-$	$+21$	-13	$+1$	-0	$+021$	-013	$+01$
13	$+013$	-3	-123	$+1$	-03	-0123	$+01$	-21	$-$	$+32$	$+2$	-021	-0	$+032$	$+02$
21	$+021$	$+2$	-1	-123	$+02$	-01	-0123	$+13$	-32	$-$	$+3$	$+013$	-032	-0	$+03$
123	-0123	-32	-13	-21	$+032$	$+013$	$+021$	$+1$	$+2$	$+3$	$-$	-01	-02	-03	$+0$
032	-32	-0123	$+03$	-02	$+123$	-3	$+2$	-0	$+021$	-013	$+01$	$+$	-21	$+13$	-1
013	-13	-03	-0123	$+01$	$+3$	$+123$	-1	-021	-0	$+032$	$+02$	$+21$	$+$	-32	-2
021	-21	$+02$	-01	-0123	-2	$+1$	$+123$	$+013$	-032	-0	$+03$	-13	$+32$	$+$	-3
0123	$+123$	-032	-013	-021	-32	-13	-21	$+01$	$+02$	$+03$	-0	$+1$	$+2$	$+3$	$-$

These are indeed the same results and are isomorphic.